

One-sided negative sectional curvature and the holomorphic Liouville property

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Abstract

We construct, in some sufficiently large fixed finite complex dimension m , a domain in $\mathbb{C}^m$ diffeomorphic to $\mathbb{R}^{2m}$ that admits a complete Kähler metric with real sectional curvature at most -1 and has only constant bounded holomorphic functions. Its sectional curvatures are unbounded below. The construction gives a negative answer to the one-sided bounded-holomorphic-function question.

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1 Introduction

For a complex manifold M , let $H^\infty(M)$ denote the algebra of bounded holomorphic functions on M . We say that M has the *holomorphic Liouville property* when $H^\infty(M) = \mathbb{C}$. The question studied here asks whether negative curvature forces this algebra to contain a nonconstant function. More precisely, the one-sided bounded-holomorphic-function question asks whether every complete simply connected Kähler manifold of complex dimension at least two satisfying

$$\text{Sec}_g \leq -a^2 < 0 \tag{1.1}$$

admits a nonconstant bounded holomorphic function. Sectional curvature here and throughout the paper means curvature on every real two-plane. There is no lower curvature bound in this hypothesis. We use the public formulation recorded by Wu and Yau [14, p. 103]; their survey [13, Section 3, p. 9] traces it to Yau's 1982 Problem 38.

Theorem 1.1. *For some finite integer $m \geq 2$ there is a domain $\mathcal{T} \subset \mathbb{C}^m$, diffeomorphic to $\mathbb{R}^{2m}$, with a complete Kähler metric g such that*

$$\text{Sec}_g \leq -1, \quad H^\infty(\mathcal{T}) = \mathbb{C}.$$

The sectional curvatures of g are unbounded below.

The dimension is chosen sufficiently large and then fixed for the entire construction. The theorem therefore disproves the universal assertion in the one-sided question. Its curvature is unbounded below, so the example leaves separate the bounded-function question under a finite two-sided bound $-b^2 \leq \text{Sec}_g \leq -a^2 < 0$.

The uniform upper bound is essential to the question. Seshadri, adapting Klembeck's radial curvature calculations, exhibited a complete Kähler metric on $\mathbb{C}^n$ with strictly negative sectional curvature [8, Section 3]. Since every bounded entire function is constant, strict negativity by itself already permits the holomorphic Liouville property. In that example some sectional curvatures tend to zero; Theorem 1.1 retains a uniform negative upper bound.

1.1 Bounded functions, bounded coordinates, and invariant metrics

A biholomorphism to a bounded domain in $\mathbb{C}^m$ supplies m bounded holomorphic coordinate functions whose differentials are independent everywhere. More generally, a bounded holomorphic map $M \rightarrow \mathbb{C}^m$ with nowhere-vanishing Jacobian provides such a tuple, without necessarily being injective. Either conclusion implies the existence of a nonconstant bounded holomorphic function. The reverse implications require additional information.

The stronger bounded-domain question appears in H. Wu's work [15, p. 195, question (1)]. Its hypotheses are completeness, simple connectivity, nonpositive real sectional curvature, and a uniformly negative upper bound for holomorphic sectional curvature. Theorem 1.1 also gives a negative answer under those hypotheses, since a negative upper bound on all real two-planes applies in particular to complex lines. Wu and Yau formulate a bounded-pseudoconvex-domain question under two-sided negative real sectional pinching in [13, Conjecture 4.3].

Comparisons among intrinsic metrics give another approach to these questions. For complete simply connected Kähler manifolds with two-sided negative real sectional pinching, Greene and Wu established a complete Bergman metric and a lower comparison with the original metric. Wu and Yau recall that result and prove the complementary upper bound [14, Theorems 4 and 6]. Under the same hypotheses, Wu and Yau [14, Corollary 7] compare the Bergman, Kobayashi–Royden,

and complete negative Kähler–Einstein metrics with the original metric. The Bergman metric uses square-integrable holomorphic top forms, and the Kobayashi–Royden metric uses holomorphic disks mapped into the manifold. After constant rescaling, bounded holomorphic functions give maps from the manifold to the disk.

These latter maps define the Carathéodory infinitesimal pseudometric: at each tangent vector, take the supremum of its lengths pulled back from the Poincaré disk metric. Every such map on $\mathcal{T}$ is constant by Theorem 1.1, so this pseudometric vanishes identically. The invariant-metric comparisons above contain no Carathéodory nondegeneracy assertion; they also require the lower curvature bound absent from our example.

Shcherbina and Zhang construct a domain in $\mathbb{C}^2$ with smooth strictly pseudoconvex boundary that is Kobayashi and Bergman complete and has the holomorphic Liouville property [9, Main Theorem and Section 6]. Their argument first forces bounded continuous plurisubharmonic functions to be constant on a Wermer-type set and then uses holomorphic uniqueness along slices. Here polynomial sampling and phases enforce bounded-function rigidity while the construction also controls sectional curvature on every real two-plane. The announced pinched result of Cao and Shaw was withdrawn [3]; it is not an input to our proof.

The separate companion *A negatively pinched Kähler threefold without bounded holomorphic coordinates* [7, Theorem 1.1] constructs a contractible domain in $\mathbb{C}^3$ with a complete two-sided negatively pinched Kähler metric and no bounded holomorphic map to $\mathbb{C}^3$ with everywhere nonzero Jacobian. That domain retains two nonconstant bounded base-coordinate functions. Its conclusion concerns bounded coordinate tuples, whereas Theorem 1.1 concerns every individual bounded holomorphic function. The two proofs are independent; no result of the companion is used below.

1.2 The construction and its main obstacles

We first separate curvature from bounded-function rigidity. Starting with a smooth strictly plurisubharmonic potential ℓ on $\mathbb{C}_s^n \times \mathbb{C}_y$, consider

$$\mathcal{T} = \{(s, y, w) : \operatorname{Im} w > \ell(s, y)\}, \quad g = \partial\bar{\partial}[-\log(\operatorname{Im} w - \ell(s, y))]. \quad (1.2)$$

The final complex dimension is $m = n + 2$: there are n spatial variables, one fiber variable y , and the tube variable w . If the potential metric of ℓ is complete and has nonpositive real sectional curvature, then the tube metric is complete and satisfies $\operatorname{Sec}_g \leq -1$. Lemma 2.1 proves this by comparing the curvature with that of complex hyperbolic space. Thus we must construct a complete nonpositively curved potential metric while placing selected large holomorphic disks in its tube.

The disks detect derivatives of bounded holomorphic functions. They lie over finite sets of spatial points obtained from a homogeneous polynomial system (H_N, p_N) of degree N , where H_N has $n - 1$ components and p_N is scalar. The grid consists of selected common zeros of H_N in projective space $\mathbb{P}^{n-1}$ at which p_N is nonzero. We use the standard unitary-invariant Gaussian polynomial ensemble and its determinant-weighted zero-density formulas, developed by Bleher, Shiffman, and Zelditch [1, Theorem 2.1 and Sections 2.2 and 2.4]. The distribution of random zeros was studied by Shiffman and Zelditch [11]; Shiffman’s later simultaneous-zero theorem [10, Corollary 1.3] includes equidistribution of the unpruned grids used here. We prove the needed special cases locally and add quantitative derivative, separation, and lift estimates.

There are on the order of N^{n-1} projective grid points. An affine lift of such a point is a representative $v \in \mathbb{C}^n$ normalized by $p_N(v) = 1$. Its N lifts are $e^{2\pi i q/N} v$, $0 \leq q < N$, and form a root-of-unity cycle. Averaging around a cycle separates Taylor degrees modulo N . In a sufficiently

large fixed dimension, the evaluation vectors of the first few surviving homogeneous terms span only a small fraction of the space of values at the grid points. Independent phases assigned to the cycles then keep a prescribed low-degree jet uniformly away from that subspace. Each testing disk keeps s fixed while varying y and w with a phase-dependent slope. Cauchy's estimate on the large disks turns this separation into exponential decay of the Taylor coefficients of $f_w = \partial f / \partial w$ for every bounded holomorphic f . Repeating each jet test at heights that grow slowly relative to the decay exponent, and applying the three-lines theorem, forces $f_w = 0$. The function then descends to a bounded entire function on $\mathbb{C}^{n+1}$ and is constant. Proposition 3.5 states the exact disk conditions separately from their realization.

The geometric obstacle is to produce those large disks without losing curvature control when their centers and directions vary. For the radial fiber we use a partial Legendre transform, in the circle-invariant bundle framework of Calabi [2, Section 3] and the momentum formulation of Hwang and Singer [5, Section 2.1]. Our profile varies over the spatial base; the additional spatial-derivative terms must be controlled for every real two-plane. The Legendre transform exposes six curvature blocks. We test their contraction on Hermitian matrices of rank at most two; every real two-plane produces a matrix in this class. This rank restriction keeps all trace losses independent of dimension. A steep fiber profile supplies positive margins in the negative curvature tensor. They absorb the spatial polynomial wells and small nonholomorphic shear coefficients, from the fiber axis to infinity.

Successive tests require different fiber centers. A direct center motion is not a holomorphic coordinate change and can destroy the near-axis estimates. We align the jets of two radial spatial metrics, temporarily widen the fiber profile, move the center while this width is available, and restore the profile with a controlled constant offset. Each period of rapid spatial growth occupies a finite interval and is followed by a capped growth regime, so no finite spatial radius is lost to blowup. Finally, every stage reserves bounds for all future wells. The local estimates therefore apply to the final potential on entire fibers and preserve every previously installed disk.

Sections 2 and 3 establish the curvature transfer and the analytic testing criterion. Section 4 derives the Legendre metric and curvature formulas. Section 5 gives the profile and shear estimates; Sections 6 and 7 supply the aligned spatial geometry and center motion. Section 8 fixes the constants before choosing the dimension, carries out the global recursion, proves completeness, and concludes Theorem 1.1.

1.3 Conventions and uniformity

We measure squared lengths using the Hermitian metric matrix: $|v|_g^2 = g_{i\bar{j}} v^i \overline{v^j}$ for the complex components of a real tangent vector. In particular a potential ℓ has metric matrix $(\ell_{i\bar{j}})$. This convention gives sectional curvatures between -4 and -1 for the ball potential $-\log(1 - |z|^2)$. For the same complex Hessian, the companion [7, Section 2] uses twice this real metric; its complex-hyperbolic sectional-curvature interval is therefore $[-2, -1/2]$. The negative Hermitian curvature tensor is denoted by $C = -R$; its coordinate formula and relation to real sectional curvature are given in Section 4.

The symbols $O(\cdot)$, $\lesssim$, and $\asymp$ denote respectively a constant-factor bound, an upper bound, and two-sided positive bounds. Dependencies are stated when the bound is introduced. Tensor jets are measured in fixed affine coordinates at the point, using operator or multilinear norms, rather than sums of their entries. Hermitian test matrices carry the Frobenius norm. For a rank at most two test T and a Hermitian form Q this gives

$$|Q : T| \leq \sqrt{2} \|Q\| |T|.$$

The constants that determine the analytic exponent gaps and the eventual dimension are independent of dimension. Constants used after the dimension has been fixed may depend on it. The order of choices is specified in Section 8; every choice respects that order of dependence.

2 The tube transfer

We first separate the passage to uniformly negative curvature from the construction of the base metric. Write $z = (z^1, \dots, z^{n+1})$ for the base coordinates and $w = u + iv$ for one additional complex coordinate. For a smooth real function ϕ , our metric convention is

$$|X|_{g_\phi}^2 = \sum_{a,b} \phi_{a\bar{b}} dz^a(X) \overline{dz^b(X)}, \quad \phi_{a\bar{b}} = \frac{\partial^2 \phi}{\partial z^a \partial \bar{z}^b}. \quad (2.1)$$

Thus the potential $|z|^2$ gives the Euclidean real metric, without an additional factor of two. We use

$$d^c \phi = -d\phi \circ J = i(\bar{\partial}\phi - \partial\phi), \quad 2\partial\phi = d\phi + i d^c \phi.$$

The real curvature tensor is normalized so that $R(X, Y, Y, X)$ is the sectional-curvature numerator and the unit sphere has positive curvature.

Lemma 2.1 (Tube transfer). *Let $\ell: \mathbb{C}^{n+1} \rightarrow \mathbb{R}$ be smooth and strictly plurisubharmonic. Suppose that its potential metric g_ℓ , in Convention (2.1), is complete and has nonpositive real sectional curvature. Set*

$$\mathcal{T}_\ell = \{(z, w) \in \mathbb{C}^{n+1} \times \mathbb{C} : b(z, w) := \operatorname{Im} w - \ell(z) > 0\}, \quad g_{\mathcal{T}} = g_{-\log b}. \quad (2.2)$$

Then $\mathcal{T}_\ell$ is connected and simply connected, indeed diffeomorphic to $\mathbb{R}^{2n+4}$, and $g_{\mathcal{T}}$ is a complete Kähler metric satisfying

$$\operatorname{Sec}_{g_{\mathcal{T}}} \leq -1.$$

No lower curvature bound on g_ℓ is required.

Proof. Topology and the metric. The map

$$(z, u, b) \mapsto (z, u + i(\ell(z) + b))$$

is a smooth diffeomorphism from $\mathbb{C}^{n+1} \times \mathbb{R} \times (0, \infty)$ onto $\mathcal{T}_\ell$. The claimed topology follows, for example by using $\log b$ as the last real coordinate.

Let $\pi(z, w) = z$. Differentiating the potential gives, as Hermitian forms,

$$g_{\mathcal{T}} = b^{-1} \pi^* g_\ell + b^{-2} \partial b \otimes \bar{\partial} b. \quad (2.3)$$

Here $d^c v = -du$, and consequently

$$2\partial b = db - i(du + d^c \ell).$$

In real notation Equation (2.3) is

$$\boxed{g_{\mathcal{T}} = b^{-1} \pi^* g_\ell + \frac{db^2 + (du + d^c \ell)^2}{4b^2}.} \quad (2.4)$$

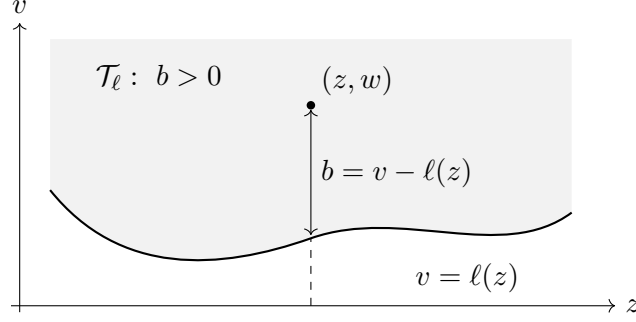


Figure 1: A schematic real slice of the tube. The coordinate $b = v - \ell(z)$ is the positive vertical gap; u and the other base directions are suppressed. No convexity of the graph is assumed.

The pullback in the first term is implicit when evaluating forms from the base. In particular, the sign before $d^c \ell$ is positive with the convention above. Formula (2.3) is positive definite: its first summand controls the base component, and on a vector with zero base component its second summand controls the remaining complex w component. Being the complex Hessian of a smooth potential, it is Kähler.

Completeness. Let γ be a locally piecewise smooth curve of finite $g_{\mathcal{T}}$ -length, with its parameter tending to an endpoint. Equation (2.4) gives

$$\frac{|db(\dot{\gamma})|}{2b} \leq |\dot{\gamma}|_{g_{\mathcal{T}}}.$$

Thus $\log b$ has finite total variation. In particular, along this curve there are constants $0 < b_- \leq b \leq b_+ < \infty$, and b has a positive limit. The same metric identity implies

$$|(\pi \circ \gamma)'|_{g_\ell} \leq \sqrt{b_+} |\dot{\gamma}|_{g_{\mathcal{T}}}.$$

The base projection therefore has finite length and converges by completeness of g_ℓ . Its image has compact closure. On that compact set $|d^c \ell|_{g_\ell}$ is bounded by some constant C , so

$$|u'| \leq |(du + d^c \ell)(\dot{\gamma})| + |d^c \ell((\pi \circ \gamma)')| \leq 2b_+ |\dot{\gamma}|_{g_{\mathcal{T}}} + C |(\pi \circ \gamma)'|_{g_\ell}.$$

Hence u also converges. Since $v = b + \ell(z)$, the curve has a limit in $\mathcal{T}_\ell$. This finite-length criterion proves metric completeness. Indeed, a Cauchy sequence without a limit would have a subsequence whose successive distances are less than 2^{-j} ; joining successive points by paths of length less than 2^{1-j} would produce a finite-length curve without a limit.

The curvature comparison. For a Hermitian matrix $g = (g_{a\bar{b}})$ arising from a Kähler potential, denote its Hermitian curvature components by

$$\mathcal{R}_{a\bar{b}c\bar{d}}^g = -\partial_a \partial_{\bar{b}} g_{c\bar{d}} + \sum_{p,q} g^{p\bar{q}} (\partial_a g_{c\bar{q}}) (\partial_{\bar{b}} g_{p\bar{d}}). \quad (2.5)$$

We record explicitly how these components determine the real sectional numerator in our normalization. If $\xi^a = dz^a(X)$ and $\eta^a = dz^a(Y)$, set

$$T^{a\bar{b}} = i(\xi^a \bar{\eta}^b - \eta^a \bar{\xi}^b).$$

Expansion of $X = \sum_a (\xi^a \partial_{z^a} + \bar{\xi}^a \partial_{\bar{z}^a})$ and similarly of Y , using the Kähler curvature symmetries, gives

$$\begin{aligned} R_g(X, Y, Y, X) &= \mathcal{R}^g(\xi, \bar{\xi}, \eta, \bar{\eta}) - \text{Re } \mathcal{R}^g(\xi, \bar{\eta}, \xi, \bar{\eta}) \\ &= \frac{1}{2} \sum_{a,b,c,d} \mathcal{R}_{a\bar{b}c\bar{d}}^g T^{a\bar{b}} T^{c\bar{d}}. \end{aligned} \quad (2.6)$$

The factor $1/2$ reflects that the complexification of the real metric satisfies $g(\partial_{z^a}, \partial_{\bar{z}^b}) = g_{a\bar{b}}/2$.

Fix a point of the tube over a base point z_0 . Choose holomorphic Kähler normal coordinates on the base, centered at z_0 , so that $(\ell_{a\bar{b}})$ is the identity and its first derivatives vanish there. Such coordinates follow by a linear normalization followed by a quadratic holomorphic change of coordinates; the Kähler symmetry makes the quadratic coefficients symmetric in their two holomorphic indices. The mixed terms of total degree three in the potential then vanish. Subtracting the real part of a holomorphic polynomial of degree at most three removes its remaining nonconstant pure terms. This subtraction is absorbed by a holomorphic translation of w : if $\ell = \ell_0 + \text{Re } H$, replace w by $w - iH$. Thus in the adapted coordinates the base potential has three-jet

$$c + |z|^2.$$

Compare the tube potential at this point with

$$\Phi_0 = -\log(\text{Im } w - c - |z|^2). \quad (2.7)$$

The two metrics and their first derivatives agree at the point. Their cubic contractions in Equation (2.5) therefore agree. In the fourth potential derivatives the only difference is the term $b^{-1}\ell_{a\bar{b}c\bar{d}}$ with all four indices in the base. The base cubic contractions vanish in the normal coordinates, so

$$\mathcal{R}^{g_T} - \mathcal{R}^{g_{\Phi_0}} = b^{-1}\pi^*\mathcal{R}^{g_\ell} \quad \text{at the comparison point.} \quad (2.8)$$

Applying Equation (2.6) gives the corresponding identity of real sectional numerators:

$$\begin{aligned} R_{g_T}(X, Y, Y, X) &= R_{g_{\Phi_0}}(X, Y, Y, X) \\ &\quad + b^{-1}R_{g_\ell}(d\pi X, d\pi Y, d\pi Y, d\pi X). \end{aligned} \quad (2.9)$$

In particular, the extra term is nonpositive for every real pair X, Y . This remains true when their base projections are linearly dependent, in which case that term is zero. The comparison metrics agree at the point, so the sectional-curvature denominators also agree.

The model normalization. Put $\tilde{w} = w - ic$. The Cayley coordinates

$$(\mathcal{Z}, \mathcal{W}) = \left(\frac{2z}{\tilde{w} + i}, \frac{\tilde{w} - i}{\tilde{w} + i} \right)$$

satisfy

$$1 - |\mathcal{Z}|^2 - |\mathcal{W}|^2 = \frac{4(\text{Im } \tilde{w} - |z|^2)}{|\tilde{w} + i|^2}.$$

Consequently the potential in Equation (2.7) equals $-\log(1 - |\mathcal{Z}|^2 - |\mathcal{W}|^2)$ up to a pluriharmonic summand and a constant. Our comparison point has $z = 0$. A real translation of $\tilde{w}$ and a dilation $(z, \tilde{w}) \mapsto (z/\sqrt{r}, \tilde{w}/r)$, $r > 0$, are model isometries and place it at $(0, i)$, which the Cayley map sends to the ball origin.

Writing Z for all the ball coordinates, expansion of $-\log(1 - |Z|^2)$ at zero gives

$$g_{a\bar{b}}(0) = \delta_{ab}, \quad \partial g_{a\bar{b}}(0) = 0, \quad \mathcal{R}_{a\bar{b}c\bar{d}}^{g_{\Phi_0}}(0) = -(\delta_{ab}\delta_{cd} + \delta_{ad}\delta_{cb}).$$

For real orthonormal X, Y at the origin, their complex components ξ, η have norm one and satisfy $\operatorname{Re}\langle \xi, \eta \rangle = 0$. Write $\langle \xi, \eta \rangle = i\beta$, where $|\beta| \leq 1$. Equation (2.6) now gives

$$\operatorname{Sec}_{g_{\Phi_0}}(X \wedge Y) = -1 - 3\beta^2 \in [-4, -1]. \quad (2.10)$$

Combining Equations (2.9) and (2.10) proves $\operatorname{Sec}_{g_T} \leq -1$ on all real planes. $\square$

The same comparison also explains why this construction need not have a lower sectional-curvature bound.

Corollary 2.2. *Under the hypotheses of Lemma 2.1, suppose that g_ℓ has a real two-plane of strictly negative sectional curvature at some point. Then*

$$\inf \operatorname{Sec}_{g_T} = -\infty.$$

Proof. Let X, Y be an orthonormal basis of such a base plane σ . At a point above it with height b , the distribution

$$\mathcal{H} = \ker db \cap \ker(du + d^c \ell)$$

is J -invariant, and $d\pi$ identifies its metric with $b^{-1}g_\ell$. If $\hat{X}, \hat{Y}$ are the horizontal lifts of X, Y , then $\sqrt{b}\hat{X}, \sqrt{b}\hat{Y}$ are g_T -orthonormal. Equation (2.9) and the model computation give the exact formula

$$\operatorname{Sec}_{g_T}(\hat{X} \wedge \hat{Y}) = -1 - 3g_\ell(JX, Y)^2 + b \operatorname{Sec}_{g_\ell}(\sigma).$$

Letting $b \rightarrow \infty$ proves the assertion. $\square$

3 Polynomial tests for bounded holomorphic functions

We will force the w -derivative of every bounded holomorphic function f on the tube to vanish. The elementary estimate comes from a holomorphic disk through a point (s_0, ζ, w_0) :

$$t \mapsto (s_0, \zeta + t, w_0 + i\alpha bt), \quad |t| < L, \quad \alpha, L > 0, \quad |b| = 1.$$

If this disk lies in the tube and $|f| \leq 1$, Cauchy's inequality gives

$$|f_y(s_0, \zeta, w_0) + i\alpha b f_w(s_0, \zeta, w_0)| \leq L^{-1}.$$

We must prevent the two derivatives from canceling at many spatial points. We arrange those points in root-of-unity cycles and hold b constant on each cycle. A finite Fourier average then keeps only Taylor degrees in one residue class. The phases will separate the prescribed lowest-degree part of f_w from the first few surviving terms of f_y , with one phase choice valid for every polynomial of that degree.

The grids constructed below provide equidistributed projective directions for polynomial sampling and cyclic affine lifts for this extraction. Their additional separation and shell estimates will be used in Section 8 to localize the shear coefficients in disjoint spatial neighborhoods when realizing the disks. After constructing the grids and choosing the phases, we formulate the precise disk conditions and prove that repeated tests force the holomorphic Liouville property.

Throughout this section, σ denotes the unitary-invariant probability measure on $\mathbb{P}^{n-1}$. All norms on vectors and derivatives are Euclidean unless indicated otherwise. Constants may depend on the fixed dimension n ; the exponents in the separation estimate below do not.

3.1 Polynomial grids and their affine lifts

Lemma 3.1 (Homogeneous polynomial grids). *Fix $n \geq 2$. For every sufficiently large integer N one can choose a system of homogeneous polynomials of degree N ,*

$$Q_N = (H_N, p_N) : \mathbb{C}^n \longrightarrow \mathbb{C}^{n-1} \times \mathbb{C},$$

with the following properties. These choices can be made along all integers $N \rightarrow \infty$, or along any prescribed sequence of degrees tending to infinity.

1. *There is a constant C_n such that*

$$N^{-C_n} \leq |Q_N(u)| \leq N \quad (|u| = 1). \quad (3.1)$$

2. *The projective zero set of H_N consists of $M_N = N^{n-1}$ transverse points. A subset $\mathcal{G}_N$ of these points satisfies*

$$|\mathcal{G}_N| = (1 - o(1))M_N, \quad \frac{1}{M_N} \sum_{\xi \in \mathcal{G}_N} \delta_\xi \rightharpoonup \sigma.$$

For every $\xi \in \mathcal{G}_N$ and every unit representative u of ξ ,

$$N^{-1} \leq |p_N(u)| \leq N, \quad \text{sing}_{\min}(DH_N(u)|_{u^\perp}) \geq N^{-1}. \quad (3.2)$$

3. *Above each $\xi \in \mathcal{G}_N$, choose any lift v_ξ such that $p_N(v_\xi) = 1$. All N lifts*

$$e^{2\pi i q/N} v_\xi, \quad 0 \leq q < N,$$

have radii in $[N^{-1/N}, N^{1/N}]$. With the absolute choice $k_ = 30$, the Euclidean balls of radius $2N^{-k_*}$ about all these lifts are pairwise disjoint. If v is one of the lifts, then*

$$|(H_N(z), p_N(z) - 1)| \geq N^{-k_*-10} \quad \text{when} \quad N^{-k_*} \leq |z - v| \leq 2N^{-k_*}. \quad (3.3)$$

In a neighborhood of radius $4N^{-k_}$ of every such lift, the first and second derivatives of Q_N have norm $O_n(N^5)$.*

In particular, weak approximation to any prescribed finite collection of continuous test functions, with any fixed positive accuracy, can be required by taking N sufficiently large.

Proof. We use the standard unitary-invariant Gaussian ensemble of homogeneous polynomials; its covariance and conditional zero-density framework appear in Bleher et al. [1, Sections 2.2 and 2.4]. The proof below supplies the needed formulas and all additional quantitative grid properties. Randomness is used only to establish existence. A standard circular complex Gaussian has density $\pi^{-1}e^{-|z|^2}$ on $\mathbb{C}$. Choose the components of Q_N independently, with

$$Q_N^{(a)}(z) = \sum_{|\beta|=N} \left(\frac{N!}{\beta!} \right)^{1/2} g_{a,\beta} z^\beta, \quad \mathbb{E}[Q_N^{(a)}(z) \overline{Q_N^{(a)}(u)}] = (z \cdot \bar{u})^N.$$

Here all $g_{a,\beta}$ are independent standard circular complex Gaussians. The covariance identity is the multinomial formula. In particular, at a unit vector the components of Q_N are independent standard complex Gaussians, and the law is invariant under unitary changes of coordinates.

We first prove (3.1) with probability tending to one. If $A = \sup_{|z|=1} |Q_N(z)|$, homogeneity gives $\sup_{|z| \leq 1+1/N} |Q_N(z)| \leq eA$. Cauchy's inequality on complex lines therefore bounds the first

derivative on the unit sphere by CNA , with an absolute constant C . A sphere net of mesh c/N , with a sufficiently small absolute c , consequently contains a point where the norm is at least $A/2$. Such a net has $O_n(N^{2n-1})$ points. Gaussian tails show

$$\mathbb{P}(A > N) \leq C'_n N^{2n-1} e^{-c'_n N^2} \longrightarrow 0.$$

On the event $A \leq N$, the Lipschitz constant on the sphere is at most CN^2 . Use a second net of mesh cN^{-C_n-2} . If the norm somewhere on the sphere is less than N^{-C_n} , it is less than $2N^{-C_n}$ at a point of this net. A standard complex Gaussian vector in $\mathbb{C}^n$ has probability at most $C'_n r^{2n}$ of lying in a ball of radius r . The union bound is therefore at most

$$C'_n N^{(C_n+2)(2n-1)-2nC_n} = C'_n N^{-C_n+4n-2}.$$

Choose, for example, $C_n = 8n$. This proves both sphere bounds with probability tending to one.

Put $d = n - 1$. Almost surely the projective zeros of the d -component system H_N are transverse. To see this directly, let $\mathcal{A}$ be its finite-dimensional complex coefficient space. Evaluation at any point of $\mathbb{P}^d$ maps $\mathcal{A}$ onto the fiber of $\mathcal{O}(N)^{\oplus d}$. Hence the universal zero incidence manifold is smooth, and a coefficient vector is a regular value of its projection to $\mathcal{A}$ precisely when the corresponding section has only transverse zeros. Sard's theorem gives a Lebesgue-null exceptional set; the Gaussian law is absolutely continuous. Transversality and compactness imply that the zero set is finite. The projective Bézout theorem for a zero-dimensional intersection of d hypersurfaces of degrees $N, \dots, N$ gives total intersection number N^d ; every transverse intersection has multiplicity one [4, Theorem III-71]. Thus the number of zeros is the deterministic quantity $M_N = N^d$ almost surely.

We next justify the counting formulas needed for equidistribution. They are the one- and two-point, zero-dimensional cases of the correlation formula in Bleher et al. [1, Theorem 2.1]. Our subsequent off-diagonal argument proves the particular convergence needed here; the more general simultaneous-zero convergence theorem of Shiffman [10, Corollary 1.3] supplies a broader context. In a holomorphic coordinate chart of $\mathbb{P}^d$, regard the system as a holomorphic map $H_a(z)$ depending linearly on its coefficients. Locally split those coefficients as (a, b) so that

$$H_{a,b}(z) = E(z)a + R(z, b), \quad \det E(z) \neq 0.$$

This is possible because evaluation is surjective. The equation $H_{a,b}(z) = 0$ is equivalent to $a = A(z, b) := -E(z)^{-1}R(z, b)$. At a zero, with $J = D_z H_{a,b}(z)$,

$$D_z A = -E(z)^{-1}J.$$

For fixed b , the change-of-variables formula with multiplicity for $z \mapsto A(z, b)$ thus introduces the real Jacobian $|\det E|^{-2}|\det J|^2$. Integrating in b against the smooth Gaussian coefficient density yields the one-point zero density

$$\rho_{1,N}(z) = p_{H_N(z)}(0) \mathbb{E}[|\det DH_N(z)|^2 \mid H_N(z) = 0]. \quad (3.4)$$

The same argument works with a bounded measurable mark multiplying the zero count. It first applies to nonnegative marks, and the general bounded case follows by decomposition.

For two distinct projective points, joint evaluation onto the two fibers is surjective. Indeed, powers of linear forms that vanish at one point and not the other give independent prescribed

values for each component. Solve now for $2d$ coefficient variables and apply the same change of variables in the two point coordinates. The point Jacobian is block diagonal, giving

$$\rho_{2,N}(z, z') = p_{(H_N(z), H_N(z'))}(0, 0) \cdot \mathbb{E}[|\det DH_N(z)|^2 |\det DH_N(z')|^2 \mid H_N(z) = H_N(z') = 0]. \quad (3.5)$$

Partitions of unity give these formulas on projective space and on the complement of its diagonal, respectively. This derivation uses only the coefficient density and ordinary change of variables; no independence of zeros has been assumed.

At a point represented by a unit vector u , take affine tangent coordinates $u + \sum_{j=1}^d z_j e_j$ with (e_j) orthonormal in $u^\perp$. Differentiating the covariance shows that $H_N(u)$ is independent of $N^{-1/2} DH_N(u)|_{u^\perp}$, and that the latter is a $d \times d$ matrix of independent standard complex Gaussians. At two distinct projective points, the cross-covariances of these normalized value and first-derivative data tend to zero locally uniformly off the diagonal. More explicitly, on a compact set of pairs with $|u \cdot \bar{v}| \leq \rho < 1$, they are bounded by a fixed power of N times ρ^{N-2} . This follows by differentiating $(u \cdot \bar{v})^N$ at most once at each point. The joint Gaussian covariance consequently converges uniformly there to the direct sum of its one-point covariances. Conditional Gaussian covariances given the values zero converge as well. Gaussian moments of the fixed polynomials $|\det J|^2$ and their products therefore converge. Equations (3.4) and (3.5) imply that the two-point density, divided by M_N^2 , converges locally uniformly off the diagonal to the product of the normalized one-point densities.

The exact total number of zeros lets us pass from off-diagonal decorrelation to weak convergence without estimating the two-point density near the diagonal. Let

$$\mu_N = \frac{1}{M_N} \sum_{H_N(\xi)=0} \delta_\xi.$$

Unitary invariance and the deterministic count imply $\mathbb{E}\mu_N = \sigma$. The positive measure $\mathbb{E}(\mu_N \otimes \mu_N)$ has total mass one. Off the diagonal, its density converges to that of $\sigma \otimes \sigma$, by the preceding calculation. This implies convergence on the whole product: given $\varepsilon > 0$, choose a compact subset of the off-diagonal region having $(\sigma \otimes \sigma)$ -mass greater than $1 - \varepsilon$. Its mass under $\mathbb{E}(\mu_N \otimes \mu_N)$ is greater than $1 - 2\varepsilon$ for all sufficiently large N . At most 2ε of the total mass can remain elsewhere. Since the diagonal has zero $(\sigma \otimes \sigma)$ -mass, there is no additional limiting mass there. Thus

$$\mathbb{E}(\mu_N \otimes \mu_N) \rightharpoonup \sigma \otimes \sigma.$$

For every continuous test function, its integral against μ_N therefore has variance tending to zero. This proves weak convergence to σ in probability.

It remains to discard the bad zeros. Use (3.4) with marks. If G is a $d \times d$ standard complex Gaussian matrix, then at a zero the derivative condition in (3.2) fails only when $\text{sing}_{\min}(G) < N^{-3/2}$. Its expected discarded fraction is

$$\frac{\mathbb{E}[|\det G|^2 \mathbf{1}_{\{\text{sing}_{\min}(G) < N^{-3/2}\}}]}{\mathbb{E}|\det G|^2} \longrightarrow 0.$$

Indeed a Gaussian square matrix is nonsingular almost surely, $|\det G|^2$ is integrable, and dominated convergence applies. The independent polynomial p_N has a standard complex Gaussian value at every unit representative. Its two forbidden tails have probability

$$\mathbb{P}(|p_N| < N^{-1}) + \mathbb{P}(|p_N| > N) = 1 - e^{-N^{-2}} + e^{-N^2} \longrightarrow 0.$$

Markov's inequality shows that the total discarded fraction tends to zero in probability. Removing that fraction does not alter weak convergence. The modulus and singular-value conditions are unchanged on multiplying a unit representative by a phase, so they hold for every such representative.

We may now select deterministic systems. Take a countable uniformly dense family of continuous functions on projective space, and impose successively accurate conditions on successively longer finite subfamilies. Convergence in probability, the vanishing bad fraction, and the probability-one transversality statement show that these conditions and (3.1) have a nonempty intersection for every sufficiently large degree. Choose one system from that intersection. A diagonal choice gives all the asserted limits and works equally well along any prescribed sequence of degrees.

We have obtained the projective sampling properties. It remains to provide disjoint neighborhoods in which Section 8 can localize its shear coefficients. Fix a good direction and a lift v with $p_N(v) = 1$. Homogeneity and (3.2) give $N^{-1/N} \leq |v| \leq N^{1/N}$. In the splitting of the domain into the radial complex line and its Hermitian orthogonal complement, $DH_N(v)$ vanishes on the radial line, while $Dp_N(v)$ has radial derivative $N/|v|$. The transverse derivative of H_N has least singular value at least N^{-4} for large N : the scaling from a unit representative costs at most one additional factor N , so this is a deliberately weaker bound than necessary. Homogeneity, the sphere upper bound, and Cauchy's inequality at scale c/N give first- and second-derivative bounds $O_n(N^5)$ in the stated neighborhood. At the lift itself, the transverse first derivatives are $O_n(N^4)$. Block inversion therefore gives

$$\|D(H_N, p_N - 1)(v)^{-1}\| \leq N^9$$

for all sufficiently large N . Taylor's formula, uniformly for $|z - v| \leq 4N^{-30}$, now yields

$$|(H_N(z), p_N(z) - 1)| \geq N^{-9}|z - v| - C'_n N^5 |z - v|^2 \geq \frac{1}{2} N^{-9} |z - v|.$$

This proves (3.3). It also excludes every other zero of $(H_N, p_N - 1)$ within distance $4N^{-30}$, proving disjointness of the balls about all good lifts. $\square$

3.2 Choosing phases uniformly for a fixed Taylor order

The grids now supply many separated evaluation points. Their next role is linear algebraic: we choose a phase at each projective point so that no fixed-order polynomial jet can be hidden among the first few Taylor degrees surviving a root-of-unity average. The independent-phase moment calculation is the same quadratic-form identity used for random-phase trace estimates in Iitaka and Ebisuzaki [6]. The additional rank and polynomial superlevel estimates give one phase choice uniformly for an entire coefficient space.

Lemma 3.2 (Phases separate low-degree jets). *Fix positive integers k, n , with $n \geq 2$, such that*

$$\sum_{j=1}^{k-1} \frac{j^{n-1}}{(n-1)!} < \frac{1}{100}. \quad (3.6)$$

Fix a nonnegative integer h . For sufficiently large polynomial grids from Lemma 3.1, choose one lift v_ξ above each good direction and let $V_N \subset \mathbb{C}^{\mathcal{G}_N}$ be the space of evaluation vectors of sums of homogeneous polynomials of degrees

$$h, h + N, \dots, h + (k-1)N.$$

Let Π_N be orthogonal projection onto V_N for the unweighted Euclidean point norm. There are phases b_ξ , $|b_\xi| = 1$, and a constant $\delta_h > 0$, independent of N and of the chosen lift phases, such that

$$M_N^{-1/2} \left\| (1 - \Pi_N)(b_\xi P(v_\xi))_{\xi \in \mathcal{G}_N} \right\| \geq \delta_h \|P\| \quad (3.7)$$

for every homogeneous polynomial P of degree h . Here $\|P\|$ is any fixed Euclidean norm on its coefficient space, chosen independently of the grid. All required grid accuracies and degree thresholds depend only on the fixed n, k, h and this norm.

Proof. Dimension gives

$$\frac{\text{rank } \Pi_N}{M_N} \leq N^{1-n} \sum_{j=0}^{k-1} \binom{h + jN + n - 1}{n - 1} < \frac{2}{100}$$

for all sufficiently large N . Indeed its limit is the sum in (3.6).

There is a number $t_h > 0$ such that, on every sufficiently accurate grid, every polynomial with $\|P\| = 1$ satisfies

$$|P(v_\xi)| > t_h \quad \text{at at least } 0.8M_N \text{ good directions.} \quad (3.8)$$

For completeness, a nonzero homogeneous polynomial has a projective zero set of σ -measure zero. In affine charts this is the elementary fact that the zero set of a nonzero complex polynomial has Lebesgue measure zero, obtained by induction and Fubini's theorem. Thus, for each norm-one polynomial P , some positive threshold has a superlevel set of measure greater than 0.9. Uniform continuity in the coefficients makes the same superlevel estimate, with a smaller threshold, valid in a neighborhood of P . The coefficient unit sphere is compact, so finitely many such neighborhoods suffice and their thresholds have a positive common lower bound. Weak convergence of the grid measures applied to these finitely many open superlevel sets gives the estimate for unit representatives, with a fixed margin in both threshold and counting proportion. Since $|v_\xi|^h \rightarrow 1$ uniformly, it gives (3.8). This proof requires only finitely many fixed approximation conditions for the given h .

The projection has $0 \leq (\Pi_N)_{\xi\xi} \leq 1$ and diagonal sum less than $0.02M_N$. Consequently at most $0.04M_N$ diagonal entries exceed $1/2$. Choose the b_ξ temporarily as independent uniform phases. Writing $Q'_N = 1 - \Pi_N$, independence gives

$$\begin{aligned} \mathbb{E} \left[M_N^{-1} \left\| Q'_N(b_\xi P(v_\xi))_\xi \right\|^2 \right] \\ = M_N^{-1} \sum_{\xi} (1 - (\Pi_N)_{\xi\xi}) |P(v_\xi)|^2 \geq 0.38t_h^2 \end{aligned}$$

when $\|P\| = 1$. The last inequality uses the intersection of the $0.8M_N$ indices in (3.8) with the complement of the at most $0.04M_N$ high-leverage indices.

We need a single phase choice for all polynomials, rather than a choice depending on P . Let (e_a) be a fixed orthonormal basis of the degree- h coefficient space, and put $q_a(\xi) = e_a(v_\xi)$. These evaluations have a common bound depending only on n, h and the chosen norm. The Gram matrix of the random linear map in (3.7) has entries

$$G_{ab} = M_N^{-1} \sum_{\xi, \eta} \overline{q_a(\xi)} \overline{b_\xi(Q'_N)_{\xi\eta}} b_\eta q_b(\eta).$$

Its diagonal-index terms are deterministic. For distinct indices, the products $\overline{b_\xi} b_\eta$ are orthonormal in L^2 of the phase probability space: two such ordered products have nonzero inner product only

when their ordered pairs agree. In particular, reversal of the pair gives expectation $\mathbb{E}(\overline{b_\xi}^2 b_\eta^2) = 0$. It follows that

$$\begin{aligned} \mathbb{E}|G_{ab} - \mathbb{E}G_{ab}|^2 &\leq C_{n,h} M_N^{-2} \sum_{\xi \neq \eta} |(\Pi_N)_{\xi\eta}|^2 \\ &\leq C_{n,h} M_N^{-2} \text{rank } \Pi_N = O_{n,h}(M_N^{-1}). \end{aligned}$$

The Gram matrix has fixed finite size. Chebyshev's inequality therefore implies that its operator-norm distance from its mean tends to zero in probability. Its mean is bounded below by $0.38t_h^2$ times the identity. A phase choice with least eigenvalue at least $0.19t_h^2$ exists for every sufficiently large grid. Taking $\delta_h = \sqrt{0.19}t_h$ proves the assertion. All estimates used only absolute values of lift evaluations, so the threshold and lower bound are uniform over their arbitrary representative phases. $\square$

Remark 3.3. For every fixed k , condition (3.6) holds in some sufficiently large finite dimension. Indeed, with $r = n - 1$, its left side is bounded by $(k - 1)^{r+1}/r!$, which tends to zero as $r \rightarrow \infty$. The argument will use one such fixed dimension, not an assertion in every dimension.

3.3 From disk estimates to the holomorphic Liouville property

Lemma 3.4 (Root-of-unity extraction). *Let $0 \leq h < N$, and suppose g is holomorphic on a neighborhood of the closed ball $|s| \leq S$, with $|g| \leq K$ there. Write $g = \sum_{r \geq 0} g_r$ for its homogeneous Taylor expansion at zero. If $R|v| < S$, then*

$$\frac{1}{N} \sum_{q=0}^{N-1} e^{-2\pi i h q/N} g(R e^{2\pi i q/N} v) = \sum_{j \geq 0} g_{h+jN}(Rv).$$

If $R|v|/S \leq e^{-0.99c/N}$ and $c \geq 1$, truncating the sum after $j = k - 1$ has error at most $CKe^{-0.99kc}$, while retaining just the term $g_h(Rv)$ has error at most $CKe^{-0.99c}$. The constant C is absolute, independent of N, h, R, S .

Proof. Cauchy's inequality on the complex line through v gives

$$|g_r(Rv)| \leq K(R|v|/S)^r.$$

The homogeneous expansion converges absolutely at the evaluation points. The finite Fourier average annihilates every degree not congruent to h modulo N , giving the identity. Put $\theta = R|v|/S$. The first tail is bounded by

$$K \sum_{j=k}^{\infty} \theta^{h+jN} \leq \frac{K e^{-0.99kc}}{1 - e^{-0.99c}}.$$

The case $k = 1$ gives the other estimate. The denominator is bounded away from zero for $c \geq 1$. $\square$

We now convert these finite-dimensional estimates into the analytic condition needed by the geometric construction. A large sheared disk bounds a linear combination of the y - and w -derivatives. The phases separate the degree- h part of the latter from the former. Heights D_j growing more slowly than the accuracy parameters c_j will let the three-lines theorem turn smallness at those heights into vanishing.

Proposition 3.5 (A testing criterion for the holomorphic Liouville property). *Fix constants $p_0, p_1 \geq 0$ and $m_0 > p_1$. Choose an integer k and a finite dimension $n \geq 2$ so that*

$$0.99k > p_0 + p_1 + 1, \quad \sum_{r=1}^{k-1} \frac{r^{n-1}}{(n-1)!} < 0.01. \quad (3.9)$$

Let (ζ_j, h_j) be a sequence of pairs from a countable dense subset of an open disk in $\mathbb{C}$ and the nonnegative integers, with every pair repeated infinitely often. At stage j , choose numbers $R_j \geq 1$, $c_j \geq 1$, an integer $N_j > h_j$, and a grid from Lemma 3.1 sufficiently large and accurate for Lemma 3.2 at order h_j . Set

$$S_j = R_j e^{c_j/N_j}, \quad \log N_j \leq c_j/100.$$

Choose heights D_j and positive numbers α_j such that

$$\alpha_j^{-1} \leq e^{p_1 c_j},$$

and, along the repetitions of every fixed pair,

$$c_j \longrightarrow \infty, \quad D_j \longrightarrow \infty, \quad \frac{D_j}{c_j} \longrightarrow 0. \quad (3.10)$$

There are phases $b_{j,\xi}$, one per good projective direction and constant throughout its cycle of lifts, with the following property.

Let ℓ be a smooth real function on $\mathbb{C}^n \times \mathbb{C}$, and let

$$\mathcal{T} = \{(s, y, w) : \operatorname{Im} w > \ell(s, y)\}.$$

Suppose every stage satisfies these conditions:

1. *On $|s| \leq S_j$,*

$$\ell(s, \zeta_j) \leq D_j - 2. \quad (3.11)$$

2. *For every holomorphic f on $\mathcal{T}$ with $|f| \leq 1$, every w with $\operatorname{Im} w = D_j$, and every $|s| = S_j$,*

$$|f_y(s, \zeta_j, w)| \leq e^{p_0 c_j}. \quad (3.12)$$

3. *For every good direction ξ , every lift $v = e^{2\pi i q/N_j} v_\xi$, and every w on that height line, the holomorphic disk*

$$t \longmapsto (R_j v, \zeta_j + t, w + i\alpha_j b_{j,\xi} t), \quad |t| < e^{m_0 c_j}, \quad (3.13)$$

is contained in $\mathcal{T}$.

Then every bounded holomorphic function on $\mathcal{T}$ is constant. The phases depend only on the grids and their assigned jet orders, not on ℓ , f , or w .

Proof. For each stage choose the phases supplied by Lemma 3.2. Assign its phase to all N_j lifts of the same direction. We prove the assertion for a holomorphic function with $|f| \leq 1$, which suffices by constant rescaling.

Fix a stage and temporarily omit its subscript. For each w with $\operatorname{Im} w = D$, put

$$A(s) = f_y(s, \zeta, w), \quad I(s) = i f_w(s, \zeta, w).$$

Both are holomorphic on a neighborhood of the closed spatial ball $|s| \leq S$, by the strict height margin and compactness. The maximum principle and (3.12) give $|A| \leq e^{p_0 c}$ on the ball. The unit w -disk centered at any (s, ζ, w) in this ball lies in $\mathcal{T}$, by (3.11); Cauchy's inequality gives $|I| \leq 1$. On a sheared disk, the derivative at $t = 0$ is $A(Rv) + \alpha b_\xi I(Rv)$. Thus

$$|A(Rv) + \alpha b_\xi I(Rv)| \leq e^{-m_0 c}. \quad (3.14)$$

Every estimate so far is uniform in the real part of w .

For each good direction take the Fourier average in Lemma 3.4 over its cycle. Since b_ξ is constant on that cycle, it factors out of the averaged second term of (3.14). The lift radius bound gives

$$\frac{R|v_\xi|}{S} \leq \exp\left(\frac{\log N - c}{N}\right) \leq e^{-0.99c/N}.$$

Let a and i' be the resulting vectors of averaged A and I values, respectively. There is a vector $a_0 \in V_N$ such that

$$\|a - a_0\|_\infty \leq C e^{(p_0 - 0.99k)c}.$$

If P is the homogeneous degree- h Taylor term of the function $s \mapsto I(Rs)$, then

$$\|i' - (P(v_\xi))_\xi\|_\infty \leq C e^{-0.99c}.$$

The averaged node inequalities still have supremum norm at most $e^{-m_0 c}$. Apply $1 - \Pi_N$ and use $a_0 \in V_N$. Since the number of good directions is at most M_N , conversion from the supremum norm to $M_N^{-1/2}$ times Euclidean norm costs at most one. Lemma 3.2 consequently gives

$$\begin{aligned} \alpha \delta_h \|P\| &\leq e^{-m_0 c} + C e^{(p_0 - 0.99k)c} + C \alpha e^{-0.99c}, \\ \|P\| &\leq C_h \left(e^{(p_1 - m_0)c} + e^{(p_1 + p_0 - 0.99k)c} + e^{-0.99c} \right). \end{aligned}$$

In particular, with the fixed positive number

$$\varepsilon = \min\{m_0 - p_1, 0.99k - p_0 - p_1, 0.99\},$$

we have

$$\|P\| \leq C_h e^{-\varepsilon c}. \quad (3.15)$$

The constants are independent of the stage once h is fixed. This is exactly the uniformity in Lemma 3.2; the Fourier tail denominator is $1 - e^{-0.99c}$, not $1 - e^{-0.99c/N}$. Since every degree- h coefficient of $I(Rs)$ is R^h times the corresponding coefficient of $I(s)$ and $R \geq 1$, (3.15) bounds each degree- h coefficient of $f_w(s, \zeta, w)$ by the same exponential rate.

We have obtained exponential decay for every coefficient of the prescribed degree, uniformly along the entire horizontal w -line. It remains to bring that estimate down from the growing testing heights to a fixed open part of the tube.

Fix now a scheduled center ζ and a multiindex β of length h . The coefficient

$$g_\beta(w) = \frac{1}{\beta!} \partial_s^\beta f_w(0, \zeta, w)$$

is holomorphic on the whole connected half-plane $\text{Im } w > \ell(0, \zeta)$. At every point of this half-plane a spatial neighborhood is available, so the derivative is well-defined and holomorphic there. Choose

once a small closed spatial polydisk about zero and a number t_0 exceeding the supremum of $\ell(s, \zeta)$ on that polydisk by more than two. Cauchy's inequalities in s and on a unit w -disk give

$$|g_\beta(w)| \leq B_\beta \quad (\text{Im } w \geq t_0),$$

where B_β is independent of every testing stage and of $\text{Re } w$. Along the stages assigned to (ζ, h) , (3.15) gives

$$\sup_{\text{Im } w = D_j} |g_\beta(w)| \leq C_h e^{-\varepsilon c_j}.$$

For any fixed $t > t_0$, apply Hadamard's three-lines theorem [12, Theorem 7.13] to the horizontal strip between heights t_0 and D_j , once $D_j > t$. Writing $\theta_j = (t - t_0)/(D_j - t_0)$, it yields

$$\sup_{\text{Im } w = t} |g_\beta(w)| \leq B_\beta^{1-\theta_j} (C_h e^{-\varepsilon c_j})^{\theta_j}.$$

If $B_\beta = 0$ there is nothing to prove. Otherwise, the logarithm of the right side is

$$\log B_\beta + \frac{t - t_0}{D_j - t_0} (\log(C_h/B_\beta) - \varepsilon c_j) \longrightarrow -\infty$$

by (3.10). Hence g_β vanishes in the upper half-plane and, by the identity theorem, throughout $\text{Im } w > \ell(0, \zeta)$. Repeating this argument for each order proves that all spatial Taylor coefficients of f_w at zero vanish at every scheduled center. Dependence of the constants and of the preliminary height t_0 on the order causes no difficulty: each order has its own infinite sequence, and analytic continuation gives the same full half-plane afterwards.

To pass from these statements to a common open set, choose a spatial ball about zero and an open subdisk of the center disk, both with compact closures. Choose L larger than the supremum of ℓ on their product. On the product with $\text{Im } w > L$, the function f_w is jointly holomorphic. At each scheduled center in the subdisk, all of its spatial Taylor coefficients vanish for every such w . Taylor uniqueness and the identity theorem on the spatial ball give vanishing on that ball. Continuity in y and density of the scheduled centers then give vanishing on the whole product. The tube is connected: the coordinates

$$(s, y, \text{Re } w, \text{Im } w - \ell(s, y))$$

identify it with $\mathbb{C}^{n+1} \times \mathbb{R} \times (0, \infty)$. The several-variable identity theorem therefore gives $f_w = 0$ everywhere on $\mathcal{T}$.

For every fixed (s, y) the w -fiber is a connected half-plane, so f is constant along that fiber. It descends to a bounded function F on $\mathbb{C}^{n+1}$. This descended function is holomorphic: near any base point choose a single constant w_0 whose imaginary part exceeds the supremum of ℓ on a small base neighborhood, and write $F(s, y) = f(s, y, w_0)$ there. These local holomorphic definitions agree on overlaps by fiber constancy. Liouville's theorem, applied on complex lines in $\mathbb{C}^{n+1}$, makes F , and therefore f , constant. $\square$

Remark 3.6. Condition (3.12) follows immediately if, at the stated sphere points and heights, the unsheared y -disk of radius $e^{-p_0 c_j}$ lies in the tube. Thus the criterion asks the geometric construction for only three quantitative facts: a height bound on a spatial ball, sufficiently wide unsheared disks on its boundary, and much wider disks in the prescribed sheared directions at its polynomial nodes. The phase choices are made before any bounded holomorphic function is considered.

4 The fiber Legendre transform and its curvature

The analytic criterion leaves us a geometric task: construct the base potential while retaining explicit curvature margins for later changes. We first derive the metric and its six curvature blocks for a fixed fiber center. The partial Legendre transform uses, up to our fixed metric normalization, the fiber momentum of the circle-invariant constructions of Calabi and Hwang–Singer [2, 5]. We allow the radial profile to vary over the spatial base, which adds spatial-derivative terms to the curvature. We compute the full tensor before estimating these additional terms.

All spatial derivatives in this section are taken with the fiber momentum τ fixed, unless a different convention is stated explicitly. The spatial variables are $s = (s_1, \dots, s_n)$, and the fiber variable is $y \in \mathbb{C}$. Its center ζ is constant in this section. Let $q(x, s) > 0$, where $x = \log \tau$, and choose F by

$$F_x = q^{-1}, \quad \lim_{x \rightarrow -\infty} (F(x, s) - x) = 0. \quad (4.1)$$

We assume that this normalization is possible, that $F \rightarrow +\infty$ as $x \rightarrow +\infty$, and that q , regarded as a function of τ , is smooth at zero with $q(0, s) = 1$. All these statements are local uniformly in s . For smooth real functions B, ψ , set

$$U(s, \tau) = B(s) + \tau \psi(s) - \int_0^\tau F(\log v, s) dv, \quad (4.2)$$

$$\rho = \log |y - \zeta|^2 = F(\log \tau, s) - \psi(s), \quad (4.3)$$

$$\ell_{\text{rad}} = U + \tau \rho = B + \int_0^\tau q(\log v, s)^{-1} dv. \quad (4.4)$$

The last identity follows by integration by parts; the boundary term $vF(\log v, s)$ tends to zero as $v \downarrow 0$. Since $F_x > 0$ and its endpoint limits are infinite with the indicated signs, (4.3) determines a unique $\tau > 0$ for every $y \neq \zeta$.

Hereafter $C = -R$ denotes the negative Hermitian curvature tensor:

$$C_{\alpha\bar{\beta}\gamma\bar{\delta}} = \partial_\alpha \partial_{\bar{\beta}} g_{\gamma\bar{\delta}} - g^{\mu\bar{\nu}} (\partial_\alpha g_{\gamma\bar{\nu}}) (\partial_{\bar{\beta}} g_{\mu\bar{\delta}}). \quad (4.5)$$

The sectional-curvature sign associated with this convention is verified below; in particular, positivity of the relevant contractions of C means nonpositive Riemannian sectional curvature.

Lemma 4.1 (Metric and regularity of the Legendre transform). *Write*

$$P = \tau q, \quad V = (B_{i\bar{j}}), \quad G = (\psi_{i\bar{j}}), \quad h = (U_{i\bar{j}}), \quad \mathcal{N} = h_\tau.$$

Where $h > 0$, the complex Hessian of ℓ_{rad} is

$$h_{i\bar{j}} ds_i d\bar{s}_j + P |d \log(y - \zeta) + U_{\tau i} ds_i|^2. \quad (4.6)$$

It is positive off the axis. Its spatial Schur complement is exactly h . The potential and its complex Hessian extend smoothly across $y = \zeta$; on that axis the Hessian in the original splitting is $V \oplus e^\psi$, and is positive whenever $V > 0$.

Proof. Use the holomorphic coordinate $z_0 = \log(y - \zeta)$, so that $\rho = z_0 + \bar{z}_0$. Differentiating $U_\tau + \rho = 0$ gives

$$U_{\tau\tau} = -P^{-1}, \quad \partial_{z_0} \tau = P, \quad \partial_{s_i} \tau = P U_{\tau i}.$$

Differentiation of $\ell = U + \tau \rho$ now gives

$$g_{0\bar{0}} = P, \quad g_{i\bar{0}} = P U_{\tau i}, \quad g_{i\bar{j}} = h_{i\bar{j}} + P U_{\tau i} U_{\tau \bar{j}},$$

which is (4.6). For regularity, write $F = \log \tau + f(\tau, s)$. The equation $f_\tau = (q^{-1} - 1)/\tau$, with $f(0, s) = 0$, shows that f is smooth. Equation (4.3) becomes $\tau e^{f(\tau, s)} = e^\psi |y - \zeta|^2$. Its derivative with respect to τ at zero is one, so the smooth implicit-function theorem applies. In particular, $\tau = e^\psi |y - \zeta|^2 + O(|y - \zeta|^4)$. Substitution in (4.4) proves the stated extension and axis metric. $\square$

We use a local holomorphic change $z_0 \mapsto z_0 + f(s)$ to make $U_{\tau i} = 0$ at a specified point. The corresponding change of U is linear in τ with a pluriharmonic spatial coefficient; it leaves h and $\mathcal{N}$ unchanged. The quadratic jet of f can also make $U_{\tau i k} = 0$ there. We call these the tilted logarithmic coordinates. The coordinate changes used to evaluate a jet are held fixed while that jet is differentiated.

Lemma 4.2 (The six curvature blocks). *In tilted logarithmic coordinates, the tensor (4.5) has the following blocks:*

$$A := C_{0\bar{0}0\bar{0}} = P^2 P_{\tau\tau} - |\partial P|_h^2, \quad (4.7)$$

$$A_i := C_{i\bar{0}0\bar{0}} = P(\partial_i P_\tau - (\mathcal{N}h^{-1}\partial P)_i), \quad (4.8)$$

$$W_{i\bar{j}} := C_{i\bar{j}0\bar{0}} = P(\partial_i \partial_{\bar{j}} \log P + P_\tau \mathcal{N}_{i\bar{j}} - P(\mathcal{N}h^{-1}\mathcal{N})_{i\bar{j}}), \quad (4.9)$$

$$C_{i\bar{0}k\bar{0}} = (\nabla_h^{2,0} P)_{ik}, \quad (4.10)$$

$$K_{ik\bar{j}} := C_{i\bar{j}k\bar{0}} = P\nabla_i^h \mathcal{N}_{k\bar{j}} + P_i \mathcal{N}_{k\bar{j}} + P_k \mathcal{N}_{i\bar{j}}, \quad (4.11)$$

$$L_{i\bar{j}k\bar{l}} := C_{i\bar{j}k\bar{l}} = C_{i\bar{j}k\bar{l}}^h + P(\mathcal{N}_{i\bar{j}} \mathcal{N}_{k\bar{l}} + \mathcal{N}_{i\bar{l}} \mathcal{N}_{k\bar{j}}). \quad (4.12)$$

Here C^h and ∇^h are the curvature with sign (4.5) and the Kähler connection of the fixed- τ spatial metric. Index positions in the products use the indicated Hermitian metric, with the usual conjugations.

Proof. In addition to the tilt, first choose spatial coordinates normal for h at the point. On functions of (s, τ) , physical coordinate derivatives act as

$$D_i = \partial_i + P U_{\tau i} \partial_\tau, \quad D_0 = P \partial_\tau,$$

and their conjugates. At the point the complete list of cubic entries needed below is

$$\ell_{00\bar{0}} = P P_\tau, \quad \ell_{i0\bar{0}} = P_i, \quad \ell_{00\bar{j}} = P_{\bar{j}}, \quad \ell_{i0\bar{j}} = P \mathcal{N}_{i\bar{j}}, \quad \ell_{ik\bar{0}} = 0, \quad \ell_{ik\bar{j}} = 0. \quad (4.13)$$

For example, the identity $P^2 \partial_\tau U_{\tau i} = P_i$, obtained by spatial differentiation of $U_{\tau\tau} = -1/P$, and its conjugate give the second and third entries. Directly differentiating the metric entries before evaluating gives

$$\begin{aligned} D_0 D_{\bar{0}} P &= P(P_\tau^2 + P P_{\tau\tau}), \\ D_i D_{\bar{0}} P &= P_i P_\tau + P \partial_i P_\tau, \\ D_i D_{\bar{j}} P &= P_{i\bar{j}} + P \mathcal{N}_{i\bar{j}} P_\tau, \\ D_i D_{\bar{j}} (P U_{\tau k}) &= P_i \mathcal{N}_{k\bar{j}} + P_k \mathcal{N}_{i\bar{j}} + P \partial_i \mathcal{N}_{k\bar{j}}, \\ D_i D_{\bar{j}} (h_{k\bar{l}} + P U_{\tau k} U_{\tau \bar{l}}) &= \partial_i \partial_{\bar{j}} h_{k\bar{l}} + P \mathcal{N}_{i\bar{j}} \mathcal{N}_{k\bar{l}} + P \mathcal{N}_{i\bar{l}} \mathcal{N}_{k\bar{j}}. \end{aligned}$$

For completeness, the fourth derivative giving the spin block has a cancellation which is useful later. Put $a_i = U_{\tau i}$. Before evaluation,

$$\ell_{ik\bar{0}} = P a_{ik} + P_i a_k + P_k a_i + P P_\tau a_i a_k.$$

At the point $a_i = a_{ik} = 0$, while

$$(a_i)_\tau = P_i/P^2, \quad (a_{ik})_\tau = P_{ik}/P^2 - 2P_iP_k/P^3.$$

Applying $D_0 = P\partial_\tau$ therefore gives P_{ik} : the two positive terms P_iP_k/P cancel the negative doubled term.

Subtract the cubic contractions in (4.5). For A the vertical contraction is PP_τ^2 and the spatial one is $|\partial P|_h^2$. For A_i they are P_iP_τ and $P(\mathcal{N}h^{-1}\partial P)_i$. For W they are $P_iP_{\bar{j}}/P$ and $P^2(\mathcal{N}h^{-1}\mathcal{N})_{i\bar{j}}$. The first combines with $P_{i\bar{j}}$ to give $P\partial_i\partial_{\bar{j}}\log P$. For the purely spatial block the contraction is the one defining C^h . These calculations prove all formulas in normal spatial coordinates. Without spatial normalization, the last entry of (4.13) is $\partial_i h_{k\bar{j}}$; its contractions replace P_{ik} by $\nabla_i^h \partial_k P$ and $\partial_i \mathcal{N}_{k\bar{j}}$ by $\nabla_i^h \mathcal{N}_{k\bar{j}}$. This also verifies the displayed tensorial formulas in general spatial coordinates. $\square$

The six blocks give the complete tensor, but the desired sign concerns real sectional curvature. The next lemma identifies the matrix tests corresponding to real two-planes and records exactly how much of the diagonal curvature can absorb each mixed block.

We specify the norm convention needed to keep all subsequent constants independent of n . The norm of a covariant tensor is its multilinear operator norm on unit vectors in the stated fixed frame. The norm of a Hermitian test matrix is its Frobenius norm. Thus, for a Hermitian matrix T of rank at most two and a covariant Hermitian form Q ,

$$|Q : T| \leq \sqrt{2}\|Q\| |T|. \quad (4.14)$$

Indeed diagonalize T ; the sum of the absolute values of its at most two nonzero eigenvalues is at most $\sqrt{2}|T|$. The same decomposition bounds a curvature pairing $C(T, T)$ by $2\|C\||T|^2$. Contractions of cubic tensors through an inverse metric can instead be regarded as inner products of two covectors. Their operator bounds therefore involve no summation factor depending on the dimension.

Lemma 4.3 (Real planes and rank-two absorption). *If $C(H, H) \geq 0$ for every Hermitian contravariant matrix H of rank at most two, the underlying real sectional curvatures are nonpositive. Write the spatial, cross, and fiber diagonal blocks of H as $T, b_\times, t$. The contraction has the terms*

$$At^2 + 2t(W : T) + 2W(b_\times, \bar{b}_\times) + L(T, T), \quad (4.15)$$

and its other terms have absolute value at most

$$2|\nabla_h^{2,0}P(b_\times, b_\times)| + 4|tA_i b_\times^i| + 4|(K : T)b_\times|. \quad (4.16)$$

In particular the following quantitative criterion suffices. Suppose $A > 0$, $W > 0$, and a nonnegative quadratic form $\mathcal{R}(T)$ satisfies

$$L(T, T) - \frac{|W : T|^2}{.98A} \geq \mathcal{R}(T), \quad (4.17)$$

$$|\nabla_h^{2,0}P(b, b)| \leq \frac{1}{4}W(b, \bar{b}), \quad (4.18)$$

$$|A_i b^i| \leq \eta\sqrt{AW(b, \bar{b})}, \quad (4.19)$$

$$|(K : T)b| \leq \eta\sqrt{\mathcal{R}(T)W(b, \bar{b})}, \quad (4.20)$$

for all such tests, where $\eta \leq 1/100$. Then

$$C(H, H) \geq .01At^2 + W(b_\times, \bar{b}_\times) + \frac{1}{2}\mathcal{R}(T). \quad (4.21)$$

All constants are independent of dimension.

Proof. For real vectors with $(1, 0)$ parts v, w , set $H = i(v \otimes \bar{w} - w \otimes \bar{v})$. This is Hermitian of rank at most two. Kähler symmetry gives

$$C(H, H) = 2\{C(v, \bar{v}, w, \bar{w}) - \text{Re } C(v, \bar{w}, v, \bar{w})\}.$$

Expand $R(v + \bar{v}, w + \bar{w}, w + \bar{w}, v + \bar{v})$. The only four nonzero terms are the two alternating type patterns and their conjugates. In the convention in which a real vector with complex components v has squared length $g_{i\bar{j}}v^i\bar{v}^j$, the complex bilinear extension of the real metric on a holomorphic and an antiholomorphic vector is one half the Hermitian pairing. Accordingly the real curvature numerator is $-\frac{1}{2}C(H, H)$. This proves the sign assertion. Expanding a general block matrix H gives (4.15)–(4.16); each mixed conjugate pair accounts for its factor two or four. A principal block T has rank at most two.

For the quantitative assertion use $2t(W : T) \geq -.98At^2 - |W : T|^2/ (.98A)$. The remaining diagonal coefficient is .02A. The spin term costs at most $W/2$. The other two terms obey

$$4\eta|t|\sqrt{AW} \leq .01At^2 + 400\eta^2W, \quad 4\eta\sqrt{\mathcal{R}W} \leq \frac{1}{2}\mathcal{R} + 8\eta^2W.$$

Since $1.5 - 408\eta^2 \geq 1$ when $\eta \leq 1/100$, (4.21) follows. $\square$

Lemma 4.4 (Exact perturbation identity). *In any fixed holomorphic coordinates let H denote the cubic jet viewed as a map from two tangent inputs to a covector, and write $H = \Gamma g$. For a perturbation with $g + \Delta g > 0$, the change of the cubic contraction $Hg^{-1}H^*$ is exactly*

$$\Gamma\Delta H^* + \Delta H\Gamma^* - \Gamma\Delta g\Gamma^* + (\Delta H - \Gamma\Delta g)(g + \Delta g)^{-1}(\Delta H - \Gamma\Delta g)^*. \quad (4.22)$$

Consequently, if $\|g^{-1}\Delta g\|$ in metric-unit axes is at most one half, the inverse of the perturbed metric is at most twice that of g , and

$$\begin{aligned} \|\Delta C\| &\leq \|\Delta j^4\ell\| + 2\|\Gamma\|\|\Delta H\| + \|\Gamma\|^2\|\Delta g\| \\ &\quad + 2\|g^{-1}\|(\|\Delta H\| + \|\Gamma\|\|\Delta g\|)^2. \end{aligned} \quad (4.23)$$

The same bound, with an absolute factor, controls rank-two contractions.

Proof. Set $Q = \Delta H - \Gamma\Delta g$. Then $H + \Delta H = \Gamma(g + \Delta g) + Q$. Multiply out the corresponding cubic contraction and subtract $\Gamma g\Gamma^*$. The two cross terms and $\Gamma\Delta g\Gamma^*$ combine to give the first three terms in (4.22). Taking operator norms gives (4.23). The inverse comparison is the elementary spectral estimate for $I + g^{-1/2}\Delta g g^{-1/2}$; the final statement uses (4.14) twice. $\square$

5 A spatially varying quiet fiber profile

We now choose a radial fiber profile that tolerates spatial variation of its parameters. Its steep part gives the curvature margins needed near the testing disks; its logarithmic tail keeps the whole affine fiber available. After establishing the scalar estimates, we use the six-block formulas to prove a uniform curvature criterion and then show that it survives localized shear terms.

Fix $b_0 \geq 1000$ and $K_0 \geq 1$; increasing K_0 later is allowed. Define a positive function u of $\tau > 0$ by

$$u = \tau \exp \left(\int_0^\tau \frac{(1+v)^{-1/16} - 1}{v} dv \right). \quad (5.1)$$

It satisfies $u/\tau \rightarrow 1$ at zero and $d \log u / d \log \tau = (1 + \tau)^{-1/16}$. It is increasing, is comparable to τ on $(0, 1]$, and tends to a finite positive limit at infinity: the last assertion follows by integrating $(1 + \tau)^{-1/16} d\tau / \tau$ from 1 to infinity. More specifically, convexity gives $(1 + v)^{-1/16} \geq 1 - v/16$, whence $e^{-1/16} \tau \leq u \leq \tau$ for $0 < \tau \leq 1$. For a real parameter E , called the severity, set

$$q = q_0(x, E) = (1 + e^E u)^{b_0} + K_0^{-1} \log(1 + \tau), \quad x = \log \tau. \quad (5.2)$$

Constants in this section may depend on b_0, K_0 , but are independent of the spatial dimension. A lower threshold for E will always be chosen sufficiently large for these constants. The primitive F has the normalization (4.1). Because $q - 1 = O_E(\tau)$ at zero, this normalization exists. Because q grows only linearly in x at the right end, $F \rightarrow +\infty$ there.

Lemma 5.1 (Scalar estimates for the profile). *Put*

$$q_a = (1 + e^E u)^{b_0}, \quad r = r_q = q_x/q, \quad D = D_q = (q_x + q_{xx})/q = r(1 + r) + r_x, \quad e_0 = q_E/q.$$

For all sufficiently large E and all $\tau > 0$,

$$q \geq 1, \quad q_x > 0, \quad D \geq r(r + \tfrac{1}{2}), \quad q_{xx}/q_x \geq -\tfrac{1}{16}, \quad (5.3)$$

$$0 < r \leq C, \quad |e_0| + |q_{EE}/q| \leq C, \quad |r_E| \leq Cr. \quad (5.4)$$

For $\tau \leq 1$, the first two quantities in the middle of (5.4) are also at most Cr . For $1 \leq j \leq 4$,

$$|\partial_E^j F| + |\partial_E^j \bar{F}| \leq C, \quad \bar{F}(\tau, E) = \tau^{-1} \int_0^\tau F(\log v, E) dv. \quad (5.5)$$

For $0 \leq j \leq 4$ and $\tau \geq e^{-E/4}$,

$$|\partial_E^j (F - \bar{F})| \leq C e^{-E/4}, \quad |\partial_E^j (F - \bar{F})| \leq Cr. \quad (5.6)$$

The following additional estimates will be used:

$$q_x \geq e^E \quad (e^{-E/4} \leq \tau \leq e^E), \quad (5.7)$$

$$q \leq C(e^{b_0 E} + \log(1 + \tau)), \quad q_x \geq c(1 + e^{b_0 E}(1 + \tau)^{-1/16}) \quad (\tau \geq 1), \quad (5.8)$$

$$r \geq .9b_0(1 + \tau)^{-1/16} \quad (1 \leq \tau \leq e^{3b_0 E}), \quad (5.9)$$

$$q \leq C, \quad q - 1 \leq C e^E \tau, \quad r \asymp e^E \tau, \quad D \asymp e^E \tau \quad (0 < \tau \leq 2e^{-E}). \quad (5.10)$$

In particular $q_x \geq c$ for $\tau \geq e^{-E}$, and $q^{-1} \leq Cr$ for $\tau \geq 2e^{-E}$.

Proof. Write $a = (1 + \tau)^{-1/16}$, $z = e^E u$, and $q_b = K_0^{-1} \log(1 + \tau)$. The logarithmic slopes of the two summands of q are

$$p = \frac{(q_a)_x}{q_a} = b_0 a \frac{z}{1 + z}, \quad d = \frac{(q_b)_x}{q_b} = \frac{\tau}{(1 + \tau) \log(1 + \tau)}.$$

They satisfy

$$p_x/p \geq -\frac{1}{16}, \quad d_x/d = \frac{1}{1+\tau} - \frac{\tau}{(1+\tau)\log(1+\tau)} \geq -\frac{1}{2}.$$

For the second inequality use $\log(1+\tau) \geq 2\tau/(2+\tau)$, obtained by differentiation from equality at zero. The derivative of a weighted mean of logarithmic slopes is the weighted mean of their derivatives plus their nonnegative variance. Hence $r_x \geq -r/2$, proving the first bound for D . Also $(q_a)_{xx}/(q_a)_x = p + p_x/p \geq -1/16$ and $(q_b)_{xx}/(q_b)_x = (1+\tau)^{-1} > 0$. Taking the weighted mean with weights $(q_a)_x, (q_b)_x$ gives the other slope bound.

Every positive E derivative of q_a of order at most four is bounded by $Cq_a \min(1, z)$. Differentiating q^{-1} thus gives

$$|\partial_E^j(q^{-1})| \leq C \frac{q_a}{q^2} \min(1, z), \quad 1 \leq j \leq 4. \quad (5.11)$$

Likewise $(q_a)_x$ and its E derivative have ratio bounded in absolute value by a profile constant, so $|r_E| \leq Cr$. Since $a \geq 2^{-1/16}$ for $\tau \leq 1$, comparison with p gives $e_0, q_{EE}/q \leq Cr$ there. The remaining bounds in (5.4) follow directly from the displayed formulas.

Differentiate the normalized integral for F . For $x \leq 0$, use $dx = d \log z/a$, where $a \geq 2^{-1/16}$, in (5.11); the resulting absolute integral is at most

$$C \int_0^\infty \frac{\min(1, z)}{(1+z)^{b_0}} \frac{dz}{z} < \infty.$$

For $x \geq 0$, put $M = e^{b_0 E}$. The positive lower and upper bounds on u for $\tau \geq 1$ imply $cM \leq q_a \leq CM$ and $q \geq cM + x/K_0$. The bound in (5.11) is therefore at most $CM/(cM + x/K_0)^2$, whose integral over $[0, \infty)$ is uniformly bounded. This proves (5.5), also for the average by integration.

Integration by parts gives, including for $j = 0$,

$$\partial_E^j(F - \bar{F}) = \int_0^1 \partial_E^j(q^{-1})(\log \tau + \log t, E) dt. \quad (5.12)$$

If $\tau \geq e^{-E/4}$, split at $t = e^{-E/4}$. On the first part the integrand is uniformly bounded. On the second part $\tau t \geq e^{-E/2}$, and $u(v) \geq c \min(v, 1)$ implies $e^E u(\tau t) \geq ce^{E/2}$. There the integrand, also for $j = 0$, is at most $Ce^{-b_0 E/2}$. This proves the first bound in (5.6). For $\tau \geq 2$, split instead at $t = \tau^{-1/2}$. Monotonicity and the positive finite limit of u , together with $\log(1 + \sqrt{\tau}) \geq \frac{1}{2} \log(1 + \tau)$, give $q(\log \sqrt{\tau}, E) \geq cq(\log \tau, E)$. Thus the integral is at most $C(\tau^{-1/2} + q^{-1})$. The estimates $q \leq C(M + \log \tau)$ and $q_x \geq c(1 + M\tau^{-1/16})$ show that $q\tau^{-1/2} \leq Cq_x$ and $1 \leq Cq_x$; hence this last bound is at most Cr . On $[e^{-E/4}, 2]$ the explicit formula for p gives a positive lower bound on r . This completes (5.6).

The estimates (5.8) follow from the same two summands of q . For $1 \leq \tau \leq e^{3b_0 E}$, the logarithmic summand divided by q_a tends uniformly to zero as the lower threshold on E increases; also $z/(1+z) \rightarrow 1$ uniformly. This proves (5.9). On $[e^{-E/4}, e^E]$ one has $z \geq ce^{3E/4}$ and $a \geq ce^{-E/16}$, so

$$q_x \geq (q_a)_x = b_0 a z (1+z)^{b_0-1} \geq c \exp((3b_0/4 - 1/16)E) \geq e^E.$$

Finally, on $(0, 2e^{-E}]$, the quantity z is comparable to $e^E \tau$ and remains bounded by a profile constant. The formulas for the first two x derivatives of q then give (5.10). Between $2e^{-E}$ and 1 the same formulas give $q_x \geq c$; for $\tau \geq 1$, use (5.8). These statements also prove the two final consequences. $\square$

The scalar estimates make two tasks compatible: the spatial Schur metric remains positive, and the fiber curvature absorbs the trace couplings in a real-plane test. The following criterion states the spatial hypotheses needed for both tasks. The word *quiet* refers to the smallness of the severity derivatives in the metric of the twist.

Proposition 5.2 (Quiet profile criterion). *There exist positive constants ϵ, μ and a lower threshold $E_{\min}$, depending on the fixed profile constants but not on n , with the following property. Let E be a smooth spatial function and use $q = q_0(x, E(s))$ with constant center and no shear. At a spatial point suppose, in some fixed holomorphic affine coordinates, that:*

1. $G \geq V > 0$. The operator norms of V, G , their inverses, their first and second metric jets, and all derivatives of E of orders one through four are at most $e^{\epsilon E}$.
2. For every Hermitian contravariant matrix T of rank at most two, $C^V(T, T) \geq e^{-\epsilon E}|T|^2$, with the norm in those affine coordinates.
3. In G -unit axes,

$$|\partial E|_G^2 \leq \mu, \quad \|\partial \bar{\partial} E\|_G \leq \mu, \quad \|\nabla_h^{2,0} E\|_G \leq \mu \quad \text{for every } \tau > 0.$$

If $E \geq E_{\min}$, the resulting metric is positive for every y over that spatial point and has nonpositive real sectional curvature there. Uniformly in τ ,

$$\mathcal{N} = (1 + O(\mu))G, \quad h \asymp V + \tau G. \quad (5.13)$$

More precisely, off the fiber axis the rank-two curvature calculation retains fixed positive multiples of

$$At^2 + W(b_\times, \bar{b}_\times) + P|T|_{\mathcal{N}}^2 + \mathbf{1}_{\{\tau \leq e^{-E/4}\}} e^{-\epsilon E}|T|^2. \quad (5.14)$$

Here the fiber blocks use the tilted logarithmic coordinates, $|T|_{\mathcal{N}}$ is its Frobenius norm in $\mathcal{N}$ -unit spatial axes, and $\mathbf{1}$ denotes the indicated indicator. The meaning of the final hypothesis is noncircular: positivity of h follows from the preceding hypotheses and does not use $\nabla_h^{2,0} E$.

Proof. Differentiating $U_\tau = \psi - F$ gives

$$\mathcal{N} = G - F_E \partial \bar{\partial} E - F_{EE} \partial E \otimes \bar{\partial} E, \quad h = V + \tau \bar{\mathcal{N}}. \quad (5.15)$$

The bar denotes averaging from 0 to τ . The bounded derivatives in (5.5) give $(1 - C\mu)G \leq \mathcal{N}, \bar{\mathcal{N}} \leq (1 + C\mu)G$. Choose μ small enough that $C\mu < 1/10$. This proves positivity and (5.13).

We will repeatedly use finite-jet estimates of the following precise kind. Differentiating (5.15) at most twice spatially gives sums with at most four derivatives of E and at most four E derivatives of F or $\bar{F}$. Each summand is a product of a fixed number of the assumed jets. The formulas for the inverse, connection, and curvature add a fixed number of inverse metric factors. Consequently there is an absolute finite constant C_* , independent of dimension, such that these estimates cost at most $Ce^{C_*\epsilon E}$, with the factors $1 + \tau$ or $(1 + \tau)^{-1}$ displayed explicitly below. This conclusion follows by the product rule and $h^{-1} \leq Ce^{\epsilon E}(1 + \tau)^{-1}I$; it uses operator norms, not sums of tensor entries. The same observation applies after normalizing spatial axes by $\mathcal{N}$, because $e^{-\epsilon E}I/C \leq \mathcal{N} \leq Ce^{\epsilon E}I$. Enlarge C_* once to cover these finitely many products, and choose $\epsilon > 0$ so small that $C_*\epsilon < 1/100$. Any profile-dependent multiplicative constants are absorbed by raising $E_{\min}$.

In $\mathcal{N}$ -unit axes define

$$\mathbf{Y} = I - \tau \mathcal{N}^{1/2} h^{-1} \mathcal{N}^{1/2}. \quad (5.16)$$

The displayed formula indicates the normalization; equivalently it is $I - \tau h^{-1}$ after $\mathcal{N} = I$ has been imposed in the fixed axes at the point. We claim

$$\|\mathbf{Y}\| \leq r/20 \quad (\tau \geq 2e^{-E}), \quad \mathbf{Y} = I + O(e^{-(1-C_*\epsilon)E}) \quad (\tau \leq 2e^{-E}). \quad (5.17)$$

Always $\|\mathbf{Y}\| \leq 1 + C\mu$, because $h \geq \tau\bar{\mathcal{N}}$ and $\bar{\mathcal{N}}$ is close to $\mathcal{N}$. On $[2e^{-E}, 1]$, the explicit slope p in the scalar lemma yields $r \geq .35b_0$ for all sufficiently large E : indeed $u \geq e^{-1/16}\tau$, $z \geq 2e^{-1/16}$, $a \geq 2^{-1/16}$, and $q_a/q \geq (1 + \log 2/K_0)^{-1} \geq (1 + \log 2)^{-1}$. The product of these three lower bounds in $p(q_a/q) = b_0a[z/(1+z)](q_a/q)$ exceeds $.35b_0$. This proves the first bound on that interval. For $\tau \geq 1$, (5.6) gives $\|\bar{\mathcal{N}} - \mathcal{N}\|_{\mathcal{N}} \leq C\mu r$. Set $h_0 = V + \tau\mathcal{N}$. The inverse identity and $h, h_0 \geq c\tau\mathcal{N}$ show that replacing h by h_0 in (5.16) costs at most $C\mu r$. In the normalized axes $V \leq \gamma I$, where $\gamma = (1 - C\mu)^{-1}$, and hence

$$\|I - \tau h_0^{-1}\| \leq \frac{\gamma}{\gamma + \tau}.$$

This is at most $r/40$ on $[1, e^{3b_0E}]$, by (5.9) and $b_0 \geq 1000$. For $\tau > e^{3b_0E}$, use $r \geq c/q$ and $q/\tau \leq C(e^{b_0E} + \log(1 + \tau))/\tau$; the supremum of the last expression on this range tends to zero with the lower threshold on E . Thus the same estimate holds there. Decreasing μ pays the inverse-identity error. Finally, if $\tau \leq 2e^{-E}$, then $h \geq V$ and the raw size bounds give $\tau\|\mathcal{N}^{1/2}h^{-1}\mathcal{N}^{1/2}\| \leq C\tau e^{C_*\epsilon E}$, proving the second bound.

Since $P_\tau = q(1 + r)$ and $P^2P_{\tau\tau} = Pq^2D$, the curvature blocks in Lemma 4.2 become

$$\frac{W}{Pq} = rI + \mathbf{Y} + q^{-1}\partial\bar{\partial}\log q, \quad (5.18)$$

$$P^{-1}\nabla_h^{2,0}P = e_0\nabla_h^{2,0}E + (q_{EE}/q)\partial E \otimes \partial E. \quad (5.19)$$

The final term in (5.18) and the spin form divided by Pq have norm at most $C\mu r$ when $\tau \geq 2e^{-E}$, and at most $C\mu$ on the remaining interval. This follows from (5.4), the three smallness assumptions on E , and $q^{-1} \leq Cr$ away from the exceptional interval. After decreasing μ ,

$$W \asymp Pqr\mathcal{N} \quad (\tau \geq 2e^{-E}), \quad W \asymp Pq(1 + r)\mathcal{N} \quad (\tau \leq 2e^{-E}), \quad (5.20)$$

and the spin inequality (4.18) holds.

For the fiber diagonal term,

$$\frac{|\partial P|_h^2}{Pq^2D} \leq C\mu \frac{e_0^2}{qD} \leq C\mu.$$

Here $h \geq c\tau G$; the final scalar quotient is bounded using $e_0 \leq Cr$ for $\tau \leq 1$, and $qD \geq cqr = cq_x \geq c$ for $\tau \geq 1$. Thus

$$A = (1 + O(\mu))Pq^2D > 0. \quad (5.21)$$

In the same normalized axes the cancellation in (4.8) gives

$$|A|_{\mathcal{N}} \leq CPq\sqrt{\mu}(|e_0|(r + \|\mathbf{Y}\|) + |r_E|) \leq CPq\sqrt{\mu}r.$$

The last inequality uses $e_0 \leq Cr$ on the exceptional interval. Combining with (5.20) and (5.21), the squared ratio to the dual norm determined by AW is at most $C\mu$. Consequently (4.19) holds with any fixed desired small $\eta > 0$ after decreasing μ .

We next pay the trace coupling. Outside the exceptional interval, (5.17) and (5.18) imply

$$|W : T| \leq Pqr \left(|\mathrm{tr}_{\mathcal{N}} T| + \frac{\sqrt{2}}{16} |T|_{\mathcal{N}} \right).$$

By $D \geq r^2$, its cost against .98A is at most

$$\frac{P}{.98(1-C\mu)} \left(|\text{tr}_{\mathcal{N}} T| + \frac{\sqrt{2}}{16} |T|_{\mathcal{N}} \right)^2. \quad (5.22)$$

For $0 \leq s \leq \sqrt{2}$, $(s + \sqrt{2}/16)^2/(1+s^2) < .8$; this can also be checked by differentiating, since its maximum on this interval occurs at the right endpoint. Because T has rank at most two, decreasing μ therefore makes (5.22) at most $.85P((\text{tr}_{\mathcal{N}} T)^2 + |T|_{\mathcal{N}}^2)$. These are exactly the positive symmetrized terms in (4.12). On $\tau \leq 2e^{-E}$, the bounds $D \geq ce^E\tau$, $q, r \leq C$, and (5.20) instead give a trace cost at most $Ce^{-E}|T|_{\mathcal{N}}^2$.

To estimate the remaining C^h , note from (5.15) that the difference $h - V = \tau\bar{\mathcal{N}}$ and its first two jets are at most $C\tau e^{C_*\epsilon E}$. On $\tau \leq e^{-E/4}$, the curvature formula or Lemma 4.4 consequently gives

$$|C^h(T, T) - C^V(T, T)| \leq C\tau e^{C_*\epsilon E} |T|^2 \leq \frac{1}{2} e^{-\epsilon E} |T|^2.$$

In the complementary range, the inverse bound and the jet bounds give

$$|C^h(T, T)| \leq C(1 + \tau) e^{C_*\epsilon E} |T|^2.$$

This is arbitrarily small compared with $P|T|_{\mathcal{N}}^2$: indeed, for $\tau \geq e^{-E/4}$, $\tau q_a/(1 + \tau) \geq e^{2E}$ after increasing the threshold on E , by the explicit power $b_0 \geq 1000$; normalization of T costs only $e^{C_*\epsilon E}$. On the exceptional interval, its trace cost $Ce^{-E}|T|_{\mathcal{N}}^2$ is likewise negligible compared with $e^{-\epsilon E}|T|^2$. We have thus obtained (4.17) with

$$\mathcal{R}(T) \geq cP|T|_{\mathcal{N}}^2 + c\mathbf{1}_{\{\tau \leq e^{-E/4}\}} e^{-\epsilon E} |T|^2, \quad (5.23)$$

before paying the K coupling.

For that coupling split K into $P\nabla^h\mathcal{N}$ and the two terms containing $P_i = Pe_0E_i$. The latter have normalized size at most $CP|e_0|\sqrt{\mu}$; their squared dual W norm after pairing with T is at most $C\mu P|T|_{\mathcal{N}}^2$. To see this directly outside the exceptional interval, the scalar quotient to bound is $e_0^2/(qr)$; it is bounded by (5.4) and (5.8). On the exceptional interval replace r in that denominator by $1 + r$, which is easier. For $P\nabla^h\mathcal{N}$ use three ranges:

1. If $\tau \leq e^{-E/4}$, then $W \geq cPq\mathcal{N}$, and the raw jet bounds give squared dual W cost at most $C\tau e^{C_*\epsilon E} |T|^2$, negligible compared with the second term of (5.23).
2. If $e^{-E/4} \leq \tau \leq e^E$, use $\|\nabla^h\mathcal{N}\|_{\mathcal{N}} \leq Ce^{C_*\epsilon E}$. The squared dual cost is at most $CPe^{C_*\epsilon E}(qr)^{-1}|T|_{\mathcal{N}}^2$. By (5.7) this is negligible compared with the first term of (5.23).
3. If $\tau > e^E$, the identity $\nabla^h h = 0$ permits

$$\nabla^h\mathcal{N} = \nabla^h(\mathcal{N} - \bar{\mathcal{N}} - V/\tau).$$

Equation (5.6), differentiated spatially once, and the raw bounds give a normalized norm at most $Ce^{-E/4+C_*\epsilon E}$. In making this estimate the connection has raw norm at most $Ce^{C_*\epsilon E}$, because h has first jets at most $C(1 + \tau)e^{C_*\epsilon E}$ and inverse norm at most $Ce^{C_*\epsilon E}/(1 + \tau)$. Since $qr = q_x \geq c$, the squared dual W cost is again negligible compared with $P|T|_{\mathcal{N}}^2$.

Fix $\eta \leq 1/100$ in Lemma 4.3. Choose μ small enough for the terms proportional to μ , and then $E_{\min}$ large enough for the exponentially small terms, so that (4.20) holds. That lemma proves positivity of every rank-two contraction with the retained margins (5.14), and hence nonpositive real sectional curvature. Smoothness and positivity at the axis follow from Lemma 4.1; curvature extends by continuity. $\square$

To realize the disks of Proposition 3.5, we need to add a term linear in the fiber variable whose coefficient varies over the spatial base. A locally constant coefficient contributes only a pluriharmonic potential. The issue is therefore its derivative support, where the estimates below must hold for arbitrarily large fiber radius.

Proposition 5.3 (Localized shear perturbations). *Under the hypotheses of Proposition 5.2, suppose E and the center are constant on the spatial region under consideration. Assume also that the first and pure second spatial derivatives of ψ , in the affine coordinates of that proposition, have norms at most $e^{\epsilon E}$. Add*

$$\ell_{\text{sh}} = \text{Re}(c_{\text{sh}}(s)(y - \zeta)).$$

On the support of its coefficient's derivatives suppose $\psi \geq -J$, where $J \geq 0$, and

$$\|j^{\leq 4} c_{\text{sh}}\| \leq e^{-CE-J/2}. \quad (5.24)$$

For a sufficiently large C depending only on the profile and the constants in the quiet criterion, this perturbation preserves metric positivity and nonpositive real sectional curvature for all y . Moreover its metric is at least one half the unperturbed metric, and fixed fractions of the curvature margins remain. The choice of C is independent of dimension.

Proof. Where the coefficient is locally constant the added potential is pluriharmonic, so it changes no metric. It remains to estimate the derivative collar in (5.24). Since E is constant, $P_i = 0$, $\mathcal{N} = G$, and $h = V + \tau G$. The holomorphic tilt used in Lemma 4.2 has linear and quadratic coefficients bounded by $Ce^{\epsilon E}$, because $U_\tau = \psi - F$. Its constant term may and will be zero. Derivatives through order four of the exponential of this fixed quadratic polynomial at the point cost at most $Ce^{C\epsilon E}$, by the ordinary product rule. These are fixed-order multilinear estimates.

First consider $\tau \geq e^{-E}$ in tilted logarithmic coordinates. The metric inverse has norm at most $Ce^{C\epsilon E}/\tau$. By (5.21), (5.20), (5.23), and $q_x \geq c$ on this range, the retained curvature on a rank-two matrix H is at least

$$c\tau e^{-C_1 E} |H|^2 \quad (5.25)$$

in these fixed coordinates, for a fixed C_1 . The cubic table (4.13) shows explicitly that the only possibly nonzero connection blocks are

$$\Gamma_{00}^0 = P_\tau, \quad \Gamma_{i0}^j = P(h^{-1}G)_i^j, \quad \Gamma_{ij}^k = (\Gamma^h)_{ij}^k.$$

Consequently their operator norm is at most $Ce^{C_1 E} q$, since r is bounded. Allowing a larger fixed power of q in the following estimates would give the same result.

The inequality $q \geq 1$ and the left normalization imply $F \leq x$. Thus

$$|y - \zeta| = e^{(F-\psi)/2} \leq e^{-\psi/2} \sqrt{\tau}.$$

After the tilt, all shear jets needed for the metric and curvature, of orders two, three, and four, are therefore at most

$$C\sqrt{\tau} \exp((-C + C_2)E), \quad (5.26)$$

where $e^{-J/2}e^{-\psi/2} \leq 1$ was used. For every fixed integer r_1 , the scalar estimate (5.8), also valid as an upper bound for small τ , gives

$$\sup_{\tau \geq e^{-E}} \frac{q^{r_1}}{\sqrt{\tau}} \leq C_{r_1} e^{C_{r_1} E}. \quad (5.27)$$

Indeed $q^{r_1} \leq C_{r_1}(e^{r_1 b_0 E} + \log(1 + \tau)^{r_1})$; on $[e^{-E}, 1]$ use $\tau^{-1/2} \leq e^{E/2}$, and on $[1, \infty)$ the quotient $\log(1 + \tau)^{r_1}/\sqrt{\tau}$ is bounded.

For clarity, this bound controls the quadratic perturbation as well as the linear terms. Write $\delta = e^{-CE}$ and absorb all other fixed E powers into e^{cE} . The relative metric error is bounded by $\delta e^{cE}/\sqrt{\tau}$. Each linear curvature error from (4.23), divided by (5.25), is bounded by $\delta e^{cE} q^{r_1}/\sqrt{\tau}$ for some fixed r_1 . The quadratic contraction error before division by that margin is at most $\delta^2 e^{cE} q^{2r_1}$; after division its bound is

$$\delta^2 e^{c'E} \left(\frac{q^{r_1}}{\sqrt{\tau}} \right)^2.$$

Equation (5.27) makes all these quantities arbitrarily small by a single sufficiently large C . It follows that the metric is at least half its old value and that the curvature errors are smaller than a fixed fraction of the retained margin, uniformly on this whole fiber range.

For $0 < \tau \leq e^{-E}$, exponentiate the tilted logarithmic coordinate and use the affine fiber coordinate with scale $e^{-(\psi(s_0)+E)/2}$ at the fixed spatial point s_0 . Denote its value by z . The zero constant term of the tilt and (5.10) give

$$|z|^2 = e^{E+F} \asymp e^{E\tau}. \quad (5.28)$$

For example, $|F - x| \leq C e^E \tau$ follows by integration of $(q^{-1} - 1)d\tau/\tau$ on this range, and proves the comparison. The metric blocks in affine coordinates are h and $P/|z|^2$. Both have inverse norm at most $e^{C_3 E}$. The curvature tensor transforms tensorially: its fiber diagonal block divides by $|z|^4$, its W block by $|z|^2$, and its spatial block is unchanged. Since

$$A \geq c\tau^2 e^E, \quad W \geq c\tau G, \quad \mathcal{R}(T) \geq c e^{-\epsilon E} |T|^2,$$

the retained rank-two curvature margin in affine coordinates is at least $e^{-C_3 E} |H|^2$, after increasing C_3 .

The connection in the new coordinates is also bounded by $e^{C_3 E}$, without an apparent singularity at $z = 0$. Indeed the logarithmic block $\Gamma_{00}^0 = P_\tau$ becomes $(P_\tau - 1)/z$, and $P_\tau - 1 = O(e^E \tau)$; thus it is bounded by $C\sqrt{e^E \tau}$. The mixed spatial block becomes $Ph^{-1}G/z$, bounded by $e^{C_3 E}$ using (5.28). The spatial-output, two-fiber-input block is zero because $P_i = 0$, and the spatial connection is already bounded. These checks cover all cubic entries. The same bounds extend to zero by the smoothness proved in Lemma 4.1.

In these affine coordinates the shear jets of orders two through four are bounded by

$$e^{-CE-J/2} e^{-(\psi(s_0)+E)/2} C e^{C_4 \epsilon E} \leq C \exp(-(C + 1/2 - C_4 \epsilon)E).$$

The estimate includes derivatives of the tilt and uses that $|z|$ is bounded on this range. Apply Lemma 4.4, with the inverse metric, connection, and reciprocal margin bounded by fixed powers of $e^{C_3 E}$. Increasing the same C makes all relative metric errors and all curvature errors smaller than one half their respective retained bounds, including at the axis. The estimates overlap at $\tau = e^{-E}$, so they cover the entire fiber.

Finally, all exponents just used arise from derivatives of order at most four, a fixed finite number of products in (4.22), and operator norms of linear maps. Rank-two contractions introduce only the absolute constants in (4.14). None of these operations counts spatial indices. This proves the asserted independence from n . $\square$

6 Radial spatial geometry and aligned profiles

The quiet criterion supplies the testing regions, where $G \geq V$. A center change requires a different arrangement: V must be close to λG with a large scalar λ . This section constructs radial spatial metrics that can pass between these arrangements and proves the curvature estimate for the aligned regime. Here *aligned* means that their metric jets through order two are close to proportional, with the scalar frozen at the evaluation point.

All norms of covariant jets in this section are multilinear operator norms in the stated fixed affine coordinates. The norm of a Hermitian contravariant matrix is its Frobenius norm. Constants called absolute may depend on a fixed upper bound for the first two derivatives of κ in $\log Y$, but not on the spatial complex dimension. This convention is essential when choosing the dimension only after the geometric constants.

6.1 The radial clocks

Write $t = |s|^2$, $\nu = \log t$, and let B and Ψ be radial potentials. Put

$$\begin{aligned} a &= B'(t), & a_\psi &= \Psi'(t) = a/\lambda, & u &= \log a, \\ Y &= \frac{du}{d\nu}, & H &= \frac{Y}{ta_\psi}, & p &= \frac{d \log a_\psi}{du}, & \kappa &= \frac{dY/d\nu}{Y(1+Y)}. \end{aligned} \quad (6.1)$$

Here a prime on a radial potential means differentiation in t . We use u for this radial variable in this section and u_s in Section 8. We write $V = \partial\bar{\partial}B$ and $G_0 = \partial\bar{\partial}\Psi$ for their Hermitian matrices.

These quantities separate the controls needed in the construction. The ratio λ compares the transverse coefficients of V and G_0 , while $H/\lambda = Y/(ta)$ is the common scale of the spatial curvature components computed below. The evolution equations will give $d\Psi/du = 1/H$ and $d\nu/du = 1/Y$: large H limits the change of the radial twist Ψ during growth, and large Y limits the spatial radius consumed. We use the first feature to prepare a center change and the second to keep each accelerated episode inside a finite radial interval. Capped growth then lets the spatial radius continue between successive tests.

Lemma 6.1 (Radial evolution). *Suppose $a, a_\psi, Y > 0$, $\lambda \geq 1$, and*

$$\frac{1}{1+Y} \leq \kappa \leq 1.1, \quad \frac{d\kappa}{d \log Y} \geq -\kappa. \quad (6.2)$$

Then

$$\begin{aligned} \frac{d\nu}{du} &= \frac{1}{Y}, & \frac{d \log Y}{du} &= \kappa(1 + 1/Y), & \frac{d \log \lambda}{du} &= 1 - p, \\ \frac{d\Psi}{du} &= \frac{1}{H}, & \frac{d \log H}{du} &= \kappa(1 + 1/Y) - p - 1/Y. \end{aligned} \quad (6.3)$$

If $Y \geq t$ initially, then $Y \geq t$ subsequently. With smooth bounded controls p, κ on each finite u interval, positive solutions have no blowup on that interval. A cap of the form $\kappa = C_\kappa/(1+Y)$, with fixed $C_\kappa \geq 1$, has no blowup at finite ν for the prescribed bounded controls.

If $p = 1$, then

$$\frac{d \log H}{d \log Y} = 1 - \frac{1}{\kappa}. \quad (6.4)$$

In particular, $p = \kappa = 1$ holds both λ and H constant. If $\kappa \geq 1$ on an interval beginning at u_0 , then the total increase of ν on that interval is at most $1/Y(u_0)$, even if its finite u length is arbitrarily large.

Proof. The first three identities follow by changing the independent variable from ν to u . Since $d\Psi/d\nu = ta_\psi$ and $H = Y/(ta_\psi)$, the fourth follows as well. Differentiating $\log H = \log Y - \nu - \log a_\psi$ gives the fifth. The lower bound on κ implies

$$\frac{d \log(Y/t)}{d\nu} = \kappa(1 + Y) - 1 \geq 0.$$

In u time, $dY/du = \kappa(1 + Y) \leq 1.1(1 + Y)$, so Y remains finite and positive on each finite interval. The remaining variables are obtained by integrating the displayed equations there. In a cap, $dY/d\nu = C_\kappa Y$, which gives the asserted global ν continuation. Dividing the last equation of (6.3), with $p = 1$, by the second proves (6.4). Finally, $\kappa \geq 1$ gives $Y(u) \geq Y(u_0)e^{u-u_0}$, and integration of $d\nu/du = 1/Y$ gives the last assertion. $\square$

The inequalities in (6.2) allow the smooth changes used later. For example, to decrease κ between two fixed positive values, make the change over a sufficiently long fixed interval in $\log Y$; then $d \log \kappa / d \log Y \geq -1$. To pass from $\kappa = 1$ to a cap, choose a smooth function $K(Y)$ which initially equals $1 + Y$, has $0 \leq K' \leq 1$, and eventually is constant, and put $\kappa = K(Y)/(1 + Y)$. The change can be made on a fixed increasing-factor interval of Y , so its first two $\log Y$ derivatives are bounded. Moreover $1 \leq K \leq 1 + Y$ and

$$Y\kappa_Y + \kappa = \frac{K + Y(1 + Y)K'}{(1 + Y)^2} \geq 0.$$

Thus all inequalities persist, and H is nonincreasing during this change when $p = 1$.

6.2 Curvature and coordinate bounds

Lemma 6.2 (Radial curvature). *Under (6.2), the negative Hermitian curvature tensor $C^V = -R^V$ satisfies*

$$C^V(T, T) \geq c\kappa \frac{H}{\lambda} |T|_V^2 \quad \text{if } T = T^*, \quad \text{rank } T \leq 2, \quad (6.5)$$

where $c > 0$ is absolute. Here the subscript V means that the matrix is expressed in V -unit axes.

Proof. At $s = (\sqrt{t}, 0, \dots, 0)$ the metric has entries

$$V_{i\bar{j}} = a\delta_{ij} + a'\bar{s}_i s_j,$$

and its radial and transverse eigenvalues are $a(1 + Y)$ and a . The unitary symmetry in the transverse variables and the Kähler symmetries leave only a transverse block, a radial-transverse bisectional block, and a pure radial entry. After removing the common factor $Y/(ta) = H/\lambda$, their respective values in V -unit axes are

$$\delta_{ij}\delta_{kl} + \delta_{il}\delta_{kj}, \quad \kappa\delta_{kl}, \quad D = \kappa(1 + \kappa + Y\kappa_Y). \quad (6.6)$$

For completeness, the transverse coefficient is $a'/a^2 = Y/(ta)$. The radial-transverse coefficient is

$$\frac{dY/d\nu}{ta(1 + Y)} = \kappa \frac{Y}{ta}.$$

For the pure radial entry, differentiate the logarithm of the radial metric along the radial complex line. Since

$$\frac{d}{d\nu} \log(a(1 + Y)) = Y(1 + \kappa),$$

its second ν derivative divided by $ta(1+Y)$ is exactly the last expression in (6.6) times $Y/(ta)$. These calculations also follow directly by subtracting the cubic contractions in the Kähler curvature formula.

Write a Hermitian test matrix in radial/transverse blocks as

$$T = \begin{pmatrix} r & b^* \\ b & S \end{pmatrix}.$$

The contraction after removing $Y/(ta)$ is

$$Dr^2 + 2\kappa r \operatorname{tr} S + 2\kappa |b|^2 + (\operatorname{tr} S)^2 + |S|^2.$$

The assumption on κ_Y gives $D \geq \kappa$. Absorb the cross term with $0.9Dr^2$. Its cost is at most

$$\frac{\kappa^2}{0.9D} (\operatorname{tr} S)^2 \leq \frac{1.1}{0.9} (\operatorname{tr} S)^2.$$

Since $\operatorname{rank} S \leq 2$, $|\operatorname{tr} S| \leq \sqrt{2}|S|$. The transverse reserve is therefore at least $5|S|^2/9$, in addition to $0.1Dr^2$ and $2\kappa|b|^2$. As $\kappa \leq 1.1$, these bound an absolute positive multiple of $\kappa(r^2 + 2|b|^2 + |S|^2)$. This proves (6.5). $\square$

Lemma 6.3 (Scaled affine jets). *Assume (6.2) and $\lambda \geq 1$. Suppose $Y \geq 1$, $Y \geq t$, the first two derivatives of κ in $\log Y$ are bounded by a fixed constant, and*

$$|p - 1| + \left| \frac{dp}{d \log Y} \right| + \left| \frac{d^2 p}{d(\log Y)^2} \right| \leq \delta, \quad 0 < \delta \leq \frac{1}{10}. \quad (6.7)$$

At a point use G_0 -unit affine axes divided by $\sqrt{H}$. In these axes $G_0 = H^{-1}I$ at the point. The transverse and radial Euclidean axis lengths are

$$\sqrt{\frac{t}{Y}}, \quad \sqrt{\frac{t}{Y(1+pY)}}. \quad (6.8)$$

In these fixed coordinates:

1. *derivatives of Ψ and B in orders one through four are bounded by C/H and $C\lambda/H$, respectively;*
2. *metric jets through order two of $V - \lambda G_0$, with λ frozen at its value at the point, are bounded by $C\delta\lambda/H$;*
3. *derivatives of u in orders one through four are bounded by C ;*
4. *if a further potential has Euclidean derivatives in orders two through four bounded by D_e , its Hessian and first two metric jets here are bounded by $CD_e/(a_\psi H)$.*

All constants are independent of dimension.

Proof. The eigenvalues of G_0 are a_ψ transversely and $a_\psi(1+pY)$ radially, giving (6.8). In these axes dt and d^2t have operator norms at most Ct/Y , and higher derivatives of t vanish. The first bound includes the fact that the only nonzero first derivative is radial; thus its axis length includes the extra factor $(1+pY)^{-1/2}$.

Set a dot to mean $d/d\nu$. We have

$$\dot{Y} = \kappa Y(1+Y) = O(Y^2), \quad \ddot{Y} = O(Y^3), \quad \ddot{\ddot{Y}} = O(Y^4).$$

The last two estimates use, respectively, the first and second indicated derivatives of κ . Also $\dot{p} = O(\delta Y)$ and $\ddot{p} = O(\delta Y^2)$. Direct differentiation gives

$$\begin{aligned} a' &= aY/t, \\ a'' &= a(Y^2 - Y + \dot{Y})/t^2, \\ a''' &= a(Y^3 - 3Y^2 + 2Y + 3(Y-1)\dot{Y} + \ddot{Y})/t^3. \end{aligned}$$

Consequently $|a^{(j)}| \leq Ca(Y/t)^j$ for $0 \leq j \leq 3$. The same formulas for a_ψ use pY in place of Y and its derivatives. They imply the corresponding estimate for a_ψ , and comparison at the point, where $a = \lambda a_\psi$, improves their difference by the factor δ .

The chain rule for a radial potential in at most four spatial slots is a finite sum of products of its t derivatives and copies of dt, d^2t . For a term containing its l th t derivative, the product of these latter factors is bounded by $C(t/Y)^l$. Since $B^{(l)} = a^{(l-1)}$, every such product is at most $Cat/Y = C\lambda/H$, and the same argument gives C/H for Ψ . The improved difference estimate proves the metric-jet assertion. For u , its t derivatives through order four are $O((Y/t)^j)$, by the displayed bounds on Y and its derivatives; the same chain rule proves the third assertion. All sums here have a number of terms bounded in terms of the derivative order, not the dimension.

Finally, the largest axis length in (6.8) is $\sqrt{t/\bar{Y}} \leq 1$. Applying two, three, or four such axes to the extra potential therefore gives at most $CD_e t/Y = CD_e/(a_\psi H)$. $\square$

6.3 Constant-center aligned estimates

In the remainder of this section the fiber center and severity E are constant, the fiber profile is $q = q_0(x, E)$ from Proposition 5.2, and no shear is added. We use the fiber notation

$$x = \log \tau, \quad P = \tau q, \quad r = q_x/q, \quad D_q = (q_x + q_{xx})/q.$$

The elementary profile estimates of Lemma 5.1 give $q \geq 1$, $r > 0$, $D_q \geq r^2$, and, for constants allowed to depend on the fixed profile and E ,

$$\begin{aligned} D_q &\geq c_E \tau \quad (0 < \tau \leq 1), & q &\leq C_E \quad (0 < \tau \leq 1), \\ q/D_q &\leq Cq^2, & q &\leq C_E(1 + \log \tau) \quad (\tau \geq 1). \end{aligned} \tag{6.9}$$

These bounds also follow directly from the positive logarithmic summand of q_0 : for $\tau \geq 1$ it supplies a positive lower bound for $q_x + q_{xx}$, whereas the other summand is bounded in τ at fixed E .

Proposition 6.4 (Aligned profile). *There is an absolute $\delta_0 > 0$ with the following property. Fix $\sigma > 0$ and take the severity E sufficiently large in terms of σ and the fixed profile. Assume the hypotheses of Lemma 6.3, with $\delta \leq \delta_0$, and also*

$$\frac{1}{2} \leq \kappa \leq 1.1, \quad H \geq \sigma, \quad G = G_0 + G_e, \quad |\partial^{[j]} G_e| \leq \delta/H \quad (0 \leq j \leq 2)$$

in its scaled affine axes. Here $G = \partial\bar{\partial}\psi$ and $\partial^{[j]}$ denotes any real or complex metric jet of order j , measured as a multilinear operator. Suppose either

$$\lambda = 1, \quad \text{or} \quad H \geq M_E(1 + \log \lambda)^2, \tag{6.10}$$

where M_E is sufficiently large depending only on the fixed profile and E . Then the constant-center fiber potential defines a positive metric with nonpositive real sectional curvature. Its spatial Schur complement satisfies $h \geq cV$.

More precisely, in scaled spatial axes and the tilted logarithmic fiber coordinate of Lemma 4.2, the contraction of its negative curvature on each Hermitian rank-at-most-two matrix with blocks (T, b, t_f) is bounded below by

$$c \left(At_f^2 + W(b, \bar{b}) + \frac{\lambda + \tau}{H} |T|^2 + P|T|_G^2 \right), \quad (6.11)$$

where $c > 0$ can be fixed independently of dimension and the stage parameters once the stated thresholds hold. The coefficient A and the form W are those of Lemma 4.2.

Proof. Since q has no spatial dependence, $\mathcal{N} = G$ and $h = V + \tau G$. Lemma 6.3 shows that, through two metric jets, h differs from the constant multiple $(\lambda + \tau)V/\lambda$ by at most $C\delta(\lambda + \tau)/H$. In particular

$$h \asymp \frac{\lambda + \tau}{H} I, \quad \Gamma^h = O(1).$$

Lemma 6.2 gives a covariant curvature lower bound $c\lambda|T|^2/H$ for V in these scaled axes. Multiplication of a metric by a positive constant multiplies its covariant curvature by that constant. The curvature difference is at most $C\delta(\lambda + \tau)|T|^2/H$: indeed its fourth-derivative term has this bound, and its cubic contractions have the same bound by the inverse estimate $|h^{-1}| \leq CH/(\lambda + \tau)$ and the first-jet bounds. Lemma 4.4 makes this calculation explicit. Reducing δ_0 therefore gives

$$C^h(T, T) \geq c \frac{\lambda + \tau}{H} |T|^2 \quad (\text{rank } T \leq 2). \quad (6.12)$$

Let $\theta = \lambda/(\lambda + \tau)$. The identity $\nabla^h(V + \tau G) = 0$ implies

$$\nabla^h G = \theta(\nabla^h G - \lambda^{-1} \nabla^h V), \quad |\nabla^h G| \leq C\delta\theta/H.$$

The curvature table gives $A = Pq^2 D_q$, vanishing spin and A_i terms, and $K = P\nabla^h G$. In G -unit normalization it gives

$$\frac{W}{Pq} = rI + \mathcal{Y}, \quad \mathcal{Y} = V_G(V_G + \tau I)^{-1}, \quad \mathcal{Y} \asymp \theta I. \quad (6.13)$$

Here V_G denotes V expressed in G -unit axes; its eigenvalues are $(1 + O(\delta))\lambda$. In particular $W > 0$. Spending a fixed small portion of W on the K coupling costs at most

$$C\delta^2 \frac{P\theta^2}{qH(r + \theta)} |T|^2 \leq C\delta^2 \frac{\tau\theta}{H} |T|^2,$$

which is an arbitrarily small portion of (6.12) after reducing δ_0 .

It remains to pay for $2t_f(W : T)$. Spending $0.98At_f^2$ costs $(W : T)^2/(0.98A)$. First suppose the second alternative in (6.10) holds. Split the two terms in (6.13) using $(a + b)^2 \leq 1.1a^2 + 11b^2$. The r term costs at most

$$\frac{1.1}{0.98} P(\text{tr}_G T)^2.$$

The horizontal curvature contains the positive term

$$P((\text{tr}_G T)^2 + |T|_G^2).$$

As $|\text{tr}_G T| \leq \sqrt{2}|T|_G$, their difference retains more than $P|T|_G^2/2$. The remaining trace cost is at most

$$C \frac{P\theta^2}{D_q H^2} |T|^2. \quad (6.14)$$

For $\tau \leq 1$, (6.9) bounds P/D_q by C_E . For $\tau \geq 1$, it bounds (6.14) by

$$\frac{C_E}{H^2} \tau \theta^2 (1 + \log \tau)^2 |T|^2 \leq \frac{C_E (1 + \log \lambda)^2}{H^2} (\lambda + \tau) |T|^2.$$

For the last inequality, put $z = \tau/\lambda$ and use boundedness of $z(1+z)^{-3}$ and $z(1+z)^{-3}(\log z)^2$ on $(0, \infty)$. Taking M_E sufficiently large absorbs this cost into a small portion of (6.12).

Now suppose $\lambda = 1$. The profile estimates give, for sufficiently large E ,

$$\|\mathcal{Y}\| \leq r/18 \quad (\tau > 2e^{-E}). \quad (6.15)$$

To see the uniformity of this assertion, $\|\mathcal{Y}\| \leq (1 + C\delta)/(1 + \tau)$. On $2e^{-E} \leq \tau \leq 1$, the first summand of q_0 gives a logarithmic slope bounded below by a large fixed multiple of one. On $1 \leq \tau \leq e^{3b_0 E}$, it gives $r \geq 0.9b_0(1 + \tau)^{-1/16}$ after increasing E . Beyond this interval the logarithmic summand gives $r \geq c/q$, which exceeds $40/(1 + \tau)$ by its linear-in-log τ upper bound and the starting size $\tau \geq e^{3b_0 E}$. These statements are unchanged by taking δ_0 smaller.

By (6.15), the trace expenditure outside the exceptional interval is at most

$$\frac{P}{0.98} \left(|\mathrm{tr}_G T| + \frac{\sqrt{2}}{18} |T|_G \right)^2.$$

For $0 \leq s \leq \sqrt{2}$, the ratio $(s + \sqrt{2}/18)^2/[0.98(1 + s^2)]$ is strictly less than one, by an absolute amount. Thus a fixed portion of $P|T|_G^2$ remains. On $\tau \leq 2e^{-E}$, the explicit left-end profile gives $q \leq C$, $r \leq C$, and $D_q \geq c\tau e^E$, where constants may depend on the fixed profile but not E . Hence the trace cost is at most $Ce^{-E}|T|^2/H^2$. Since $H \geq \sigma$, this is a small portion of $(1 + \tau)|T|^2/H$ for E sufficiently large in terms of σ .

All cases retain fixed portions of A , W , (6.12), and the symmetrized PG^2 term, proving (6.11). Lemma 4.3 gives nonpositive real sectional curvature off the fiber axis. The metric is positive there and $h \geq cV$. Smoothness of the original affine fiber potential and its positive limiting metric extend both conclusions across the axis. $\square$

7 Changing the fiber center

A moving center is not a holomorphic change of coordinates. We therefore first widen the fibers, then estimate the actual change of potential as a perturbation in fixed holomorphic coordinates. The widening and the metric scales must be chosen in the order specified below.

Throughout this section $p = \kappa = 1$ and $Y \geq \max\{1, t\}$, so λ, H are constant by Lemma 6.1, and $V = \lambda G_0$ as metrics, including their jets. We use the scaled affine spatial coordinates of Lemma 6.3. The severity E is constant, and no shear term is present. The old twist has the form

$$\psi_{\mathrm{old}} = \Psi + \psi_e, \quad G_e = \partial \bar{\partial} \psi_e, \quad |\partial^{[j]} G_e| \leq \delta/H \quad (0 \leq j \leq 2),$$

with δ sufficiently small as in Proposition 6.4. Suppose also that, throughout the interval being configured,

$$\psi_{\mathrm{old}} \leq A_{\mathrm{old}}, \quad |\partial^{[j]} \psi_{\mathrm{old}}| \leq A_{\mathrm{old}} \quad (1 \leq j \leq 4) \quad (7.1)$$

in these fixed scaled axes at each point. The number A_{old} is known independently of the target sizes of λ, H . Additive constants already placed in the twist are included in this upper bound.

Proposition 7.1 (Center change). *Choose the profile constants in the following order: first the fixed exponent b_0 of the quiet profile; then a sufficiently large absolute constant d_0 ; and then K_0 sufficiently large in terms of d_0 and absolute constants specified in the proof. Fix E and old data as in (7.1). There is a width parameter L , chosen using only these data and the fixed profile, such that the following construction is possible.*

After choosing L , choose a sufficiently large λ , and then a sufficiently large H ; explicitly one can require

$$\lambda \geq \max\{e^{3L}, \Lambda_f\}, \quad H \geq M_f(1 + \log \lambda)^2, \quad (7.2)$$

where Λ_f, M_f are finite constants determined by the fixed profile, E, L , the old data, and δ , but not by λ, H . Over three fixed units of the variable Ψ , one can change from any prescribed constant center in the unit disk to any other such center, returning to the original profile $q_0(x, E)$ at the end. The final twist differs from the old twist by a nonnegative constant, possibly very large.

During the change the potential is smooth, its metric is positive and has nonpositive real sectional curvature, and the metric bounds cV from below on spatial projection, for a fixed $c > 0$. There is no restriction on the cost of the final constant offset. All power exponents and smallness tolerances used before choosing the spatial dimension are dimension independent.

7.1 The profile deformation and its elementary bounds

Write $F_x^0 = 1/q_0$ and normalize $F^0(x) - x \rightarrow 0$ at the left end. We record properties of the concrete quiet profile that will be used here. At fixed E ,

$$\begin{aligned} q_0 &\geq 1, \quad (q_0)_x > 0, \quad (q_0)_{xx}/(q_0)_x \geq -1/16, \\ |(q_0)_x| + |(q_0)_{xx}| &\leq Cq_0, \quad q_0(x, E) \leq C_E + C(1 + \max(x, 0)), \\ \inf_{x \geq -20} (q_0)_x &> 0. \end{aligned} \quad (7.3)$$

Here and below a subscript E permits dependence on E and the already fixed profile. The last infimum can be bounded below using just the derivative of $K_0^{-1} \log(1 + e^x)$. The inequality for the second derivative follows by differentiating both summands in the quiet profile. In particular $(q_0)_x + (q_0)_{xx} \geq (15/16)(q_0)_x$.

Two endpoint properties are also needed. The function $Q_0(v) = q_0(\log v, E)$ is smooth down to $v = 0$, with

$$Q_0(0) = 1, \quad Q'_0(0) = b_0 e^E + K_0^{-1} > 0. \quad (7.4)$$

At the other end the auxiliary function in the quiet profile tends to a positive finite limit with error $O(e^{-x/16})$, so

$$q_0(x, E) = c_E + x/K_0 + O_E(e^{-x/16}) \quad (x \rightarrow \infty).$$

Integration, using $1/q_0 - K_0/x = O_E(x^{-2})$, gives

$$F^0(x) = K_0 \log x + O_E(1), \quad \int_0^L \frac{dx}{q_0(x, E)} = K_0 \log L + O_E(1). \quad (7.5)$$

No uniformity in E is asserted or needed in these endpoint constants; E has been fixed before L .

Take L large. Choose a smooth function χ with

$$0 \leq \chi \leq 1, \quad \text{supp } \chi \subset [-20, 3L], \quad \chi = 1 \text{ on } [0, L], \quad \int \chi = 2L, \quad \chi' \geq -1/16,$$

and with its derivatives through any fixed needed order bounded absolutely. Such a function is obtained from a fixed rising cutoff, an adjustable plateau, and a fixed descending cutoff of length greater than 16; adjusting the plateau gives the integral exactly. Define the increasing inverse map $m = m(x)$ by

$$x = m + L - \frac{1}{2} \int_{-\infty}^m \chi(v) dv. \quad (7.6)$$

Then

$$1 \leq m' \leq 2, \quad m \leq x, \quad m = x - L \quad (x \leq L - 20), \quad m = x \quad (x \geq X := 3L). \quad (7.7)$$

Its derivatives of any fixed order are bounded absolutely. Set

$$q_1(x) = q_0(m(x), E) \leq q_0(x, E).$$

Since $m'' = \chi'(m)(m')^3/2$, putting $a = m' \in [1, 2]$ gives

$$\begin{aligned} (q_1)_x + (q_1)_{xx} &\geq (q_0)_x(m)(a - a^2/16 - a^3/32) \\ &\geq \frac{3}{4}(q_0)_x(m)a = \frac{3}{4}(q_1)_x > 0. \end{aligned} \quad (7.8)$$

Also $|(q_1)_x| + |(q_1)_{xx}| \leq Cq_1$, with constants independent of L . These bounds follow from (7.3) and the bounds on m', m'' .

7.2 Offsets during profile interpolation

On the first unit of Ψ , choose a smooth nondecreasing flat-ended cutoff $\beta = \beta(\Psi)$ from 0 to 1, with fixed bounds on its derivatives through order four. Define

$$q(x, \beta) = (1 - \beta)q_0(x, E) + \beta q_1(x), \quad \mathfrak{s} = 1/q, \quad (7.9)$$

and let F^β have derivative $F_x^\beta = \mathfrak{s}$ and left normalization $F^\beta - x \rightarrow 0$. Every such primitive exists, tends to infinity on the right, and differs from F^0 by a constant for $x \geq X$. All spatial derivatives in the rest of this subsection are at fixed $\tau = e^x$.

Use an offset $w^+(\beta)$ defined by $w^+(0) = 0$ and

$$(w^+)'(\beta) = \partial_\beta F^\beta(X) + \frac{d_0}{1 - \beta + 1/L}, \quad D(x, \beta) = (w^+)'(\beta) - \partial_\beta F^\beta(x). \quad (7.10)$$

Here the prime on w^+ denotes differentiation in β . Since $\mathfrak{s}_\beta \geq 0$ and vanishes for $x \geq X$,

$$D \geq \frac{d_0}{1 - \beta + 1/L}. \quad (7.11)$$

This lower bound can dominate $\sup_x |\mathfrak{s}_\beta|$ by any required fixed factor if d_0 is sufficiently large. Indeed

$$0 \leq \mathfrak{s}_\beta = (q_0 - q_1)/q^2 \leq \min\{(1 - \beta)^{-1}, \sup_{x \leq X} q_0(x, E)\}.$$

By taking L sufficiently large relative to C_E , the second entry is at most CL with an absolute C . The elementary bound $\min\{z^{-1}, CL\} \leq (C + 1)/(z + 1/L)$ proves the assertion, including $\beta = 1$ by continuity.

For later restoration use, on another unit of Ψ , the reverse mixture $q = (1 - \beta)q_1 + \beta q_0$ with a fresh increasing cutoff. Now $\mathfrak{s}_\beta \leq 0$ and $\partial_\beta F^\beta \leq 0$. Choose a new offset $w^-(\beta)$ with $w^-(0) = 0$

and constant positive derivative greater than a sufficiently large fixed multiple of $1 + \sup_{x,\beta} |\mathfrak{s}_\beta|$. With $D = (w^-)' - \partial_\beta F^\beta$, the same domination holds. No estimate on the size of $w^-(1)$ is required.

Once L is fixed, all the following constants are finite and may be included in a number C_f : upper bounds for the offsets and their derivatives through order four; for D and its first three β derivatives; for the profile and the finite-order derivatives used below on $x \leq X$; for their left-end decay coefficients; and the reciprocal lower constants in the positive estimates on that interval. To check finiteness at the noncompact left endpoint, both $q_0 - 1$ and $q_1 - 1$, with their needed derivatives, are $O_f(e^x)$. Thus β derivatives of $\mathfrak{s}$ are $O_f(e^x)$, and their integrals defining derivatives of F^β converge. In particular C_f may include e^X and large functions of E, L . It is selected before λ, H .

Lemma 7.2 (Curvature during interpolation). *During either profile switch above, keep the center constant and set $\psi = \psi_{\text{old}} + w^\pm(\beta(\Psi))$, also including the already frozen offset when making the reverse switch. If λ and then H satisfy sufficiently large thresholds of the form (7.2), the metric is positive, $h \geq cV$, and its negative curvature has a strictly positive rank-two reserve off the fiber axis. These conclusions extend across that axis by continuity.*

Proof. A prime on β now means differentiation in Ψ . Direct differentiation of $\psi - F^\beta$ gives

$$\mathcal{N} = (1 + D\beta')G_0 + G_e + (D\beta'' + D_\beta(\beta')^2)\partial\Psi \otimes \bar{\partial}\Psi. \quad (7.12)$$

Here D_β is the derivative at fixed x ; thus all dependence of the primitive on the spatial cutoff is included. The last term and its first two spatial jets are $O(C_f/H^2)$, because all positive derivatives of Ψ in scaled axes are $O(1/H)$. The derivatives of the scalar $D\beta'$ beyond its value are $O(C_f/H)$. With $d_f = 1 + D\beta' \geq 1$, these facts show

$$\mathcal{N} \asymp d_f G_0 > 0, \quad h = V + \int_0^\tau \mathcal{N}(v) dv \geq cV.$$

The constants in $\asymp$ can be absolute by making H large enough; the value d_f itself is bounded by C_f .

First consider $x \leq X$. Since $\tau \leq e^X$, the first two metric jets of $h - V$ are $O(C_f/H)$. Hence

$$\Gamma^h = \Gamma^{G_0} + O(C_f/\lambda), \quad C^h(T, T) \geq c\lambda|T|^2/H \quad (\text{rank } T \leq 2). \quad (7.13)$$

The second estimate follows from Lemma 6.2, the scaled jet estimates, and the curvature perturbation identity, taking $\lambda \gg C_f$.

The mixed form in Lemma 4.2 is

$$\frac{W}{Pq} = (1 + r)\mathcal{N} - \tau\mathcal{N}h^{-1}\mathcal{N} + q^{-1}\partial\bar{\partial}\log q.$$

The last term equals

$$-\mathfrak{s}_\beta\beta'G_0 + O(C_f/H^2).$$

The spin form, divided by Pq , equals

$$-\mathfrak{s}_\beta\beta'\nabla_h^{2,0}\Psi + O(C_f/H^2).$$

In fact $q_\beta/q^2 = -\mathfrak{s}_\beta$; the remaining terms contain two first derivatives of Ψ . Lemma 6.3 and (7.13) give $|\nabla_h^{2,0}\Psi| \leq C/H$ with absolute C . Also $|\tau\mathcal{N}h^{-1}\mathcal{N}| \leq C_f/(\lambda H)$. Choose d_0 first so that the

domination in (7.11) is sufficiently strong in comparison with this absolute C , and then choose λ, H large compared with C_f . It follows that

$$W \geq c \frac{Pq}{H} d_f I, \quad |\nabla_h^{2,0} P(b, b)| \leq \frac{1}{4} W(b, \bar{b}). \quad (7.14)$$

This argument also applies to the reverse mixture, using absolute values of $\mathfrak{s}_\beta$.

By (7.8) and convexity of the mixture,

$$D_q = (q_x + q_{xx})/q \geq c_f \min(1, \tau) \quad (x \leq X).$$

Left-end decay and boundedness on the rest of this interval give

$$|\partial \log P| + |\partial P_\tau| \leq C_f \min(1, \tau)/H.$$

It follows from the curvature table that

$$A = (1 + o(1)) P q^2 D_q, \quad |A_i| \leq C_f P \min(1, \tau)/H, \quad (7.15)$$

where the relative error tends uniformly to zero as λ, H increase. To verify uniformity at $\tau = 0$, the subtracted term in A is at most $C_f P^2 \min(1, \tau)^2 / (\lambda H)$, while the leading term is $P q^2 D_q$. The ratio tends to zero uniformly under the stated choices. Similarly (7.14) and (7.15) make the A_i coupling arbitrarily small in units $(AW)^{1/2}$.

Finally $|K/P| \leq C_f/H$. Spending fixed portions of A, W according to Lemma 4.3 costs at most

$$C_f \frac{P}{qH} |T|^2 \leq \frac{C_f}{H} |T|^2 \quad \text{for the } K \text{ coupling}, \quad C_f \frac{P}{D_q H^2} |T|^2 \leq \frac{C_f}{H^2} |T|^2 \quad \text{for the trace coupling}.$$

In these displayed bounds C_f is enlarged to include e^X ; also P/D_q is bounded on $x \leq X$, including the left endpoint. Both costs are small compared with (7.13). Thus fixed fractions of A, W , and $\lambda |T|^2/H$ remain.

Now let $x > X$. Since the two profiles coincide exactly here, $F^\beta(x) - F^0(x)$ is independent of x . Consequently $\mathcal{N} = \mathcal{N}_\infty$ is independent of τ , and

$$h = V_f + \tau \mathcal{N}_\infty, \quad V_f = V + \int_0^{e^X} (\mathcal{N}(v) - \mathcal{N}_\infty) dv. \quad (7.16)$$

There is no infinite tail in this expression. Its first two spatial jets differ from those of V by $O(C_f/H)$. Treating $d_f = 1 + D\beta'$ at the point as constant, the jets of $\mathcal{N}_\infty$ differ from those of $d_f G_0$ by at most $C\delta d_f/H$, after taking $H \gg C_f/\delta$; similarly those of V_f differ from λG_0 by a sufficiently small relative amount after taking $\lambda \gg C_f/\delta$.

The aligned argument can now be applied with the two spatial scales λ and d_f . More explicitly, let $\theta_f = \lambda/(\lambda + \tau d_f)$. The comparisons and the identity $\nabla^h(V_f + \tau \mathcal{N}_\infty) = 0$ give

$$C^h(T, T) \geq c \frac{\lambda + \tau d_f}{H} |T|^2, \quad |\nabla^h \mathcal{N}_\infty| \leq C\delta d_f \theta_f / H.$$

The K cost is bounded by $C\delta^2(\lambda + \tau d_f)|T|^2/H$. The rI part of the trace cost uses the same rank-two symmetrized $P\mathcal{N}_\infty^2$ reserve as before. The remaining cost is at most

$$C \frac{\tau q d_f^2 \theta_f^2}{D_q H^2} |T|^2 \leq C_f \frac{(1 + \log \lambda)^2 (\lambda + \tau d_f)}{H^2} |T|^2.$$

For the last step, use (6.9) and the scalar inequality from the proof of Proposition 6.4 with λ/d_f in place of λ . We have $1 \leq d_f \leq C_f$ and may require $\lambda \geq C_f$, so the logarithm is bounded by $\log \lambda$. Taking $H \geq M_f(1 + \log \lambda)^2$ with sufficiently large M_f absorbs this last cost. Spin and A_i vanish in this region because $q = q_0$ is spatially constant.

We have proved positivity and nonpositive curvature off the axis, with fixed fractional reserves. The profiles are smooth in τ at zero, equal to one there, and have the prescribed left normalization. Thus the potential and metric are smooth in the original affine fiber coordinate, whose metric at the axis has entries V, e^ψ . Curvature extends by continuity. This proves the lemma. $\square$

7.3 The width gained before moving the center

At the end of the first switch freeze $\beta = 1$ and the offset, and write $\psi = \psi_{\text{old}} + w^+(1)$. Changing variables in the defining integral gives

$$F^1(x) = L + F^0(m(x)) - \frac{1}{2} \int_{-\infty}^{m(x)} \frac{\chi(v)}{q_0(v, E)} dv. \quad (7.17)$$

Both sides have derivative $1/q_1$ and the same left normalization. At $x = X$, $m(X) = X$, so integration of (7.10) yields

$$w^+(1) = L - \frac{1}{2} \int_{\mathbb{R}} \frac{\chi(v)}{q_0(v, E)} dv + d_0 \log(1 + L). \quad (7.18)$$

In particular this offset is nonnegative: the integral is at most $\int \chi = 2L$.

Define the pointwise width scales

$$R_0 = \exp((L - \psi)/2), \quad Z = \exp((F^1(x) - \psi)/2) = |y - \zeta|.$$

Because $\chi = 1$ on $[0, L]$, (7.5) and (7.18) imply

$$R_0 \geq c_{E, A_{\text{old}}} L^{K_0/4 - d_0/2}. \quad (7.19)$$

Choose K_0 so large relative to d_0 that $K_0/4 - d_0/2 > K_0/9$. Then, for L sufficiently large depending on old data,

$$R_0 \geq L^{K_0/9}, \quad Z \geq x^{K_0/9} \quad (x \geq L - 20). \quad (7.20)$$

Here is the second bound in detail. At $x = L - 20$, formula (7.17) gives $F^1 = L + F^0(-20)$, so Z is a fixed old-data multiple of R_0 . Monotonicity of F^1 and (7.19) prove the desired bound on $[L - 20, 3L]$, absorbing the fixed factor $3^{K_0/9}$ by increasing L . For $x \geq 3L$, the integral in (7.17) has reached its total and cancels the integral in (7.18). Thus exactly

$$F^1(x) - \psi = F^0(x) - \psi_{\text{old}} - d_0 \log(1 + L).$$

The asymptotic (7.5), together with $x \geq 3L$ and the strict exponent inequality already chosen, gives the second bound on this remaining interval. These estimates are uniform over the spatial interval because only the upper bound in (7.1) was used.

7.4 The actual nonholomorphic center motion

The preceding calculation has produced width before any center is moved. We now spend that width to control the metric and curvature errors of moving the center. Two coordinate systems cover the fiber: logarithmic coordinates away from the enlarged core and a scaled affine coordinate on the core, including its axis.

On the next unit of Ψ , hold $q = q_1$ and all offsets fixed, and let ζ be a smooth flat-ended convex-combination interpolation of the two prescribed centers. Its derivatives through order four as a function of Ψ are bounded absolutely. By Lemma 6.3, and $H \geq 1$, its spatial derivatives in scaled axes satisfy

$$|\partial^{[j]}\zeta| \leq C/H \quad (1 \leq j \leq 4). \quad (7.21)$$

At a point s_0 , compare the moving-center potential with the potential obtained by replacing $\zeta(s)$ by the constant $\zeta(s_0)$ in a neighborhood. This is a comparison in fixed holomorphic coordinates, not a claim that $y - \zeta(s)$ is a holomorphic coordinate. All constants in the following two comparisons may depend on old data, E , and the fixed profile, but are independent of L, λ, H once the stated thresholds hold.

First suppose $x \geq L - 20$. For the frozen center use scaled spatial axes and the tilted logarithmic fiber coordinate of Lemma 4.2. The previous interpolation proof with cutoff constant retains fractions of A, W and the horizontal curvature reserve. On this region $m \geq -20$, so (7.3) and (7.8) give

$$A \geq c\tau, \quad W \geq c\tau H^{-1}I.$$

The horizontal reserve is at least $c\tau/H$: on $x \leq X$ use $\lambda \geq e^X \geq \tau$ in (7.13), and on $x > X$ use the aligned reserve. Consequently the full rank-two curvature contraction is bounded below by $c\tau/H$ times the squared Frobenius norm of the test. The metric has the same lower scale and its inverse has norm at most CH/τ . These assertions also follow directly for the metric from $h = V + \tau G$ and $P = \tau q$.

The polynomial holomorphic tilt has coefficients bounded by the first two old twist derivatives; the offset is constant. In these tilted coordinates the frozen connection has norm at most Cq . Indeed the possibly large entries in the cubic table are $P_\tau = q + q_x = O(q)$ and $P\mathcal{N}h^{-1} = O(q)$, whereas $P_i = 0$, the tilted pure holomorphic second spatial derivative vanishes, and $\Gamma^h = O(1)$.

Let Φ be the one-variable fiber potential as a function of $\rho + \psi = \log|y - \zeta|^2 + \psi$. Its first derivative is τ , and differentiation in this argument is $q\partial_x$. Thus its derivatives of orders one through four have size at most $C\tau q^3$. For example

$$\Phi'' = \tau q, \quad \Phi''' = \tau q(q + q_x), \quad \Phi'''' = \tau q((q + q_x)^2 + q(q_x + q_{xx})).$$

The fixed argument has bounded jets through order four in the chosen frame. The difference in its argument after moving the center is

$$\log \left| 1 - \frac{\zeta(s) - \zeta(s_0)}{y - \zeta(s_0)} \right|^2. \quad (7.22)$$

It vanishes at the evaluation point, and each positive jet contains a derivative of ζ and a denominator of size at least Z . By (7.21), bounded tilt coefficients, and $HZ \geq 1$, its jets through order four are $O(1/(HZ))$. The chain rule consequently bounds every potential-jet difference through order four by

$$C \frac{\tau q^3}{HZ}. \quad (7.23)$$

In particular its relative metric error is at most Cq^3/Z . Once this is small, Lemma 4.4, the bound $\Gamma = O(q)$, and the inverse bound $O(H/\tau)$ show that the curvature error on unit rank-two tests is at most

$$C \frac{\tau}{H} \left(\frac{q^5}{Z} + \frac{q^8}{Z^2} \right) \leq C \frac{\tau q^8}{HZ}. \quad (7.24)$$

The first term accounts for all linear fourth-jet and cubic-contraction changes, and the second is the quadratic inverse-contraction term. Fixed-rank contraction costs only an absolute factor.

For $x \geq L - 20$, (7.3) and $m \leq x$ give $q = q_1 \leq Cx$ once L is large relative to C_E . This constant is independent of L, λ, H . Choose, in addition to the earlier requirement, $K_0/9 > 10$. Then (7.20) makes the ratios in (7.23) and (7.24) arbitrarily small compared with the metric and curvature reserves by increasing L . This choice precedes λ, H .

It remains to include $x < L - 20$ and the axis itself. Put $v = \tau e^{-L}$. By (7.7), the fiber potential added to B is exactly e^L times the original smooth radial function, now evaluated at

$$|y - \zeta(s)|^2 e^{\psi(s) - L}.$$

This identity follows by changing variables in $\int_0^\tau q_1(\log t)^{-1} dt$. Use scaled spatial axes and the fixed affine fiber coordinate

$$z = \frac{y - \zeta(s_0)}{R_0(s_0)}.$$

At the evaluation point $|z|^2 = e^{F^0(\log v)} \asymp_E v$ for $0 \leq v \leq e^{-20}$. The comparison is uniform down to zero because $F^0(\log v) - \log v$ is smooth there. The smooth radial function and its derivatives on this fixed compact argument interval have bounds depending only on old data and E .

For the frozen center, (7.4) gives, with positive old-data constants,

$$A \geq c_E \tau^2 e^{-L}, \quad W \geq c_E \tau H^{-1} I,$$

throughout $0 < v \leq e^{-20}$. For the first inequality, the continuous positive ratio of $P^2 P_{\tau\tau}$ to $\tau^2 e^{-L}$ has a positive limit at zero by $Q'_0(0) > 0$; positivity away from zero follows from (7.3). For W , the term $Pq\mathcal{V}G$ has a positive lower bound of this size because $\tau \leq e^{L-20}$ and $\lambda \geq e^{3L}$. Divide the vertical and cross curvature blocks by $|z|^4$ and $|z|^2$, respectively, on changing from logarithmic to affine fiber coordinates. The retained vertical and cross reserves are then at least $c_E e^L$ and $c_E e^L/H$. The horizontal reserve $c\lambda/H$ also dominates ce^L/H . The metric has the same smallest scale, so its inverse is bounded by $C_E H/e^L$.

Undoing the exponentiated tilt changes the affine fiber variable by $z \mapsto e^{\theta(s)} z$, where $\theta(s_0) = 0$ and its coefficients are bounded by old twist derivatives. Since $|z|$ is bounded, the differential and its inverse, and the needed higher jets, have old-data bounds independent of L, λ, H . This preserves the indicated rank-two lower margin up to old-data constants. The connections in these coordinates are bounded: the logarithmic entry transforms to

$$(P_\tau - 1)/z = O_E(|z|),$$

and the spatial-fiber entry is bounded by

$$|P\mathcal{N}h^{-1}|/|z| \leq C_E \tau / (\lambda|z|) \leq C_E |z|,$$

using $\lambda \geq e^L$. The pure vertical-to-spatial entry vanishes before the bounded tilt is undone. The remaining spatial entries are bounded by the radial jet estimates.

In this affine frame the moving smooth argument, before applying the fixed smooth radial function, is

$$\left| z - \frac{\zeta(s) - \zeta(s_0)}{R_0(s_0)} \right|^2 e^{\psi(s) - \psi(s_0)}. \quad (7.25)$$

Its difference from the frozen argument has jets through order four bounded by $C/(HR_0)$, including at $z = 0$, by (7.21). Therefore the potential jets change by at most $Ce^L/(HR_0)$. The inverse metric and bounded connection estimates give a curvature error at most

$$C \frac{e^L}{H} (R_0^{-1} + R_0^{-2}). \quad (7.26)$$

By (7.20), this is a small fraction of the e^L/H reserve for L sufficiently large, independently of the later choices of λ, H . The metric error is small by the same estimate. Smoothness includes the axis, so no limiting logarithmic-coordinate assertion is being used at a singular coordinate point.

We have proved that the moving-center metric remains, say, at least half the frozen-center metric and that its real sectional curvature is nonpositive. Its spatial projection bound is therefore inherited from the frozen Schur bound $h \geq cV$.

Completion of Proposition 7.1. Choose d_0 to dominate the absolute spin constants in the interpolation estimate, and choose K_0 so that $K_0/4 - d_0/2 > K_0/9$ and $K_0/9 > 10$. The latter explicit exponent is larger than the exponent 8 in (7.24). For fixed old data and E , choose L sufficiently large for (7.20) and for the moving-center metric and curvature errors to be less than the retained reserves. Next compute the finite constants C_f for both switches and choose λ and then H according to (7.2), with the larger thresholds needed in Lemma 7.2. This is the required noncircular order of choices.

Perform the forward switch with constant old center, the center interpolation with frozen profile q_1 and offset, and finally the reverse switch with constant new center. Lemma 7.2 controls the two switches and the preceding argument controls the actual center motion. The cutoffs are flat-ended, so all joins are smooth. At the end the profile is q_0 , the center is the prescribed new constant, and the added offset is $w^+(1) + w^-(1) \geq 0$. All profiles are at least one, have smooth left endpoints with value one, and have normalized primitives tending to infinity on the right; thus they define smooth fiber potentials for all affine fiber values throughout. The spatial projection bound follows on each part, with a fixed fractional loss during the center motion. No bound on the final constant offset is required or asserted. $\square$

8 Assembly of the complete base

The local estimates are now available. We must realize infinitely many analytic tests with one finite spatial dimension, preserve the earlier tests when adding later wells, and obtain a complete global metric. We choose the spatial data and join the local regimes. All tensor and fixed-order derivative estimates in this section use operator or multilinear norms in fixed affine coordinates. In particular, their leading exponents do not include a trace over the spatial dimension.

Proposition 8.1. *For some fixed finite integer n , there is a smooth strictly plurisubharmonic potential ℓ on $\mathbb{C}^n \times \mathbb{C}$ whose Kähler metric is complete and has nonpositive real sectional curvature. The potential satisfies every hypothesis of Proposition 3.5, for a countable dense set of centers in the unit disk and every spatial Taylor order. Thus all bounded holomorphic functions on its half-plane tube are constant.*

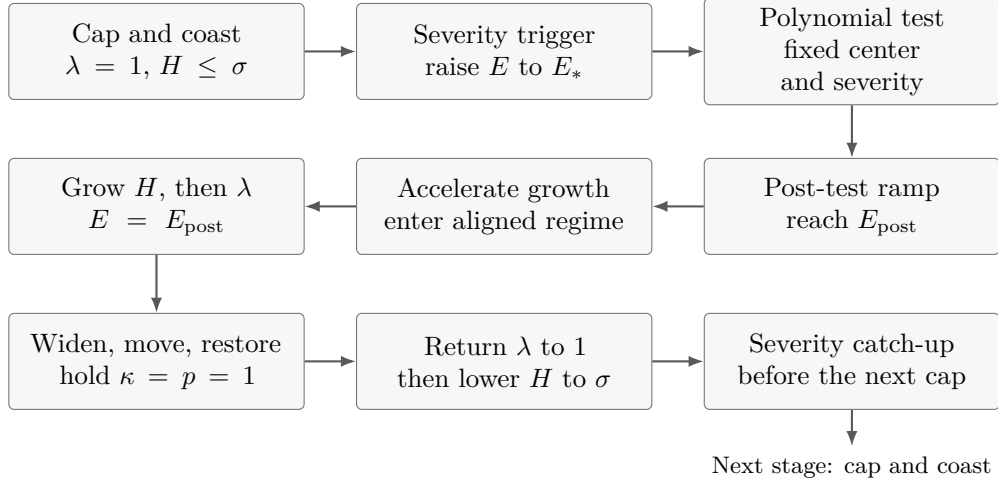


Figure 2: One stage, in increasing spatial radius. The test is installed while the spatial growth is capped. The intervening growth and center change occupy finite clock intervals. After the reset, severity must catch up with the accumulated spatial scale before the next capped regime can begin. The arrows indicate order, not elapsed radius.

At each spatial point the fiber potential is recovered from

$$F(\log \tau, s) = \log |y - \zeta(s)|^2 + \psi(s)$$

with the left normalization of Section 4. The radial function B supplies the spatial metric. The twist ψ starts with a radial background Ψ : polynomial wells make selected fibers wide, and additional radial plurisubharmonic terms control the derivatives of the severity. A term $\text{Re}(c_{\text{sh}}(s)(y - \zeta(s)))$ prescribes the sheared disk directions. We choose these data in increasing spatial radius. The polynomial wells are the only later additions that extend into earlier regions, so their inward tails must be reserved in advance.

The proof occupies this section. A cycle starts on a capped spatial interval, raises the severity, performs one test, and raises severity again on a post-test ramp. At constant severity it then accelerates the spatial metric, enters the aligned regimes, changes center, and restores the spatial parameters. Finally it increases severity while the normalized spatial geometry is fixed, until the severity can pay for the next capped interval.

We retain the radial notation of Lemma 6.1:

$$t = |s|^2, \quad \nu = \log t, \quad a = B'(t), \quad u_s = \log a, \\ Y = \frac{du_s}{d\nu}, \quad a_\psi = \Psi'(t) = a/\lambda, \quad H = \frac{Y}{ta_\psi}, \quad p = \frac{d \log a_\psi}{du_s}.$$

Here H is a scalar; the polynomial system below is denoted by $\mathcal{H}$. The evolution equations are

$$\frac{dY}{du_s} = \kappa(1 + Y), \quad \frac{d\nu}{du_s} = Y^{-1}, \quad \frac{d \log \lambda}{du_s} = 1 - p, \quad (8.1)$$

$$\frac{d\Psi}{du_s} = H^{-1}, \quad \frac{d \log H}{du_s} = \kappa(1 + Y^{-1}) - p - Y^{-1}. \quad (8.2)$$

The admissibility and derivative conditions on κ, p in the radial lemmas will always be imposed.

8.1 Wells and the order of constants

Fix a sequence (ζ_j, h_j) in which every pair from a fixed countable dense subset of $\{|y| < 1\}$ and $\mathbb{Z}_{\geq 0}$ occurs infinitely often. The parameters of a test will be

$$E_* = B_* c, \quad \eta = e^{-c}, \quad N = \lfloor e^{\gamma E_*} \rfloor, \quad l = \frac{N}{2} \log(t/R^2), \quad S = R e^{c/N}. \quad (8.3)$$

The constants B_*, γ are fixed below. At each stage R is fixed before c . For a degree- N system $(\mathcal{H}, p_N)$ from Lemma 3.1, define

$$W_j(s) = d \log \frac{|\mathcal{H}(s/R)|^2 + |p_N(s/R) - 1|^2 + e^{-2c}}{1 + e^{-2c}}. \quad (8.4)$$

The polynomial p_N is unrelated to the radial control p . The logarithmic sum-of-squares formula shows that W_j is smooth and plurisubharmonic.

Lemma 8.2. *There is an absolute exponent C_{pol} such that, for $t \geq 1$ and $1 \leq r \leq 4$,*

$$|\partial^{[r]} W_j| \leq C(n, d) \exp\{C_{\text{pol}}(\max(l, 0) + c + \log N)\}. \quad (8.5)$$

There is a constant $L_n < \infty$ such that, after increasing the degree threshold,

$$|\partial^{[r]} W_j| \leq N^{L_n} \quad (1 \leq r \leq 4, \quad l \geq L_n \log N). \quad (8.6)$$

Both statements hold throughout the indicated outer ranges.

Proof. Homogeneity and the spherical upper bound give $|(\mathcal{H}, p_N)(s/R)| \leq N e^l$. Cauchy estimates at scale $c_0 \sqrt{t}/N$ bound its derivatives of order at most four by this quantity times a fixed power of N , since $t \geq 1$. In a differentiated logarithm there are only a fixed number of factors and denominator powers; the denominator in (8.4) is at least e^{-2c} . This proves (8.5) with an absolute exponent; fixed-dimensional constants and d affect only its prefactor.

The spherical lower bound also gives $|(\mathcal{H}, p_N)(s/R)| \geq e^l N^{-C_n}$. Once l exceeds a sufficiently large multiple of $\log N$, subtraction of $(0, 1)$ leaves norm at least $e^l N^{-C_n}/2$. The factors e^l in derivatives of the logarithm can then be canceled. The remaining costs are fixed powers of N , with exponents depending on the already fixed n . Increase L_n to pay them and the degree threshold to pay the prefactor. $\square$

For each stage we also introduce smooth nonnegative radial plurisubharmonic functions $Z_{j,\text{in}}$ and $Z_{j,\text{out}}$. They vanish before their stage and eventually become affine nondecreasing functions of ν . The center changes add offsets that vanish before their change interval and are constant afterwards. We seek the final twist in the form

$$\psi = \Psi + \sum_j W_j + \sum_j (Z_{j,\text{in}} + Z_{j,\text{out}}) + \text{center-change offsets}. \quad (8.7)$$

The center is constant except on its change intervals. There is one fiber profile at each spatial point: it is $q_0(x, E(s))$ except during the coordinated switches of Proposition 7.1. Shears occur only in the disjoint test collars. The recursion below will make the well sum converge smoothly on compact spatial sets; the other added terms are locally finite.

The constants must provide the three outputs of Proposition 3.5: a fixed-height spatial ball, small unsheared disks on its boundary, and large prescribed sheared disks at the nodes. Their

exponents determine k and hence the dimension. Accordingly we first choose all geometric constants independently of n , and only then use stagewise size choices to pay dimensional prefactors.

We give the full order of choices. First choose the profile constants, including d_0, K_0 for the center change, then ϵ, μ for Proposition 5.2, the shear exponent in Proposition 5.3, and the tolerance δ for Proposition 6.4. These choices are independent of n . Use control transitions whose first two derivatives in $\log Y$ have absolute bounds. Let C_{rad} dominate the absolute exponents needed for radial raw jets through order four. Choose

$$\epsilon_1 \ll \epsilon/(1 + C_{\text{rad}}), \quad C_{\uparrow} \gg (1 + C_{\text{rad}})/\epsilon. \quad (8.8)$$

Choose next

$$m_z \geq 10(C_{\text{pol}} + 1), \quad B_* \geq \frac{C_*(1 + m_z + C_{\text{pol}} + C_{\text{rad}})}{\epsilon_1}, \quad (8.9)$$

where the absolute C_* is large enough for the finitely many raw exponent comparisons below. Choose C_s larger than the quiet shear severity exponent, and C_w large enough to pay the shell loss and four derivatives of its cutoffs; $C_w > 5k_* + 20$ suffices for these latter costs after increasing the other fixed constants if necessary. Here k_* is the absolute separation exponent of the polynomial grids. Choose

$$0 < \gamma \ll \epsilon_1, \quad \gamma B_* \leq \min\{.005, (8C_w)^{-1}\}, \quad k_b > \max\{8/\gamma, 100/\epsilon\}. \quad (8.10)$$

Choose M_b sufficiently large for the ratios in Lemma 8.4, and then $\sigma > 0$ sufficiently small for that lemma and Lemma 8.7. These are choices made from fixed constants; in particular $\sigma M_b k_b^2 \ll 1$, and σ pays all quiet smallness requirements involving $C_{\uparrow}$. Severity thresholds can depend on σ . Finally take

$$d \geq 8(M_b k_b + C_s + 1)B_*. \quad (8.11)$$

All constants so far are independent of dimension.

Set

$$m_0 = d/2, \quad p_0 = 2d, \quad p_1 = C_s B_* + C_w d \gamma B_*. \quad (8.12)$$

Then $p_1 \leq d/4 < m_0$. Choose an integer k with $.99k > p_0 + p_1 + 1$, and only now fix a finite n sufficiently large that

$$\sum_{r=1}^{k-1} \frac{r^{n-1}}{(n-1)!} < .01. \quad (8.13)$$

Such a finite n exists because k is fixed. The constants C_n, L_n and all dimensional prefactors will be paid by the stagewise choice of large c ; they do not change (8.12).

8.2 Recursive convention and capped intervals

Every estimate for an already configured region is made with an additional allowance of one for the absolute value and spatial jets through order four of all future wells. Let ν_j^{sat} denote the designated endpoint of severity saturation at stage j , and let

$$K_j = \{s : |s|^2 \leq e^{\nu_j^{\text{sat}}}\}$$

be the entire inner spatial ball ending there. The testing annulus lies beyond this ball. At stage j we impose

$$\sup_{s \in K_j} |\partial^{[r]} W_j(s)| \leq 2^{-j}, \quad 0 \leq r \leq j + 4. \quad (8.14)$$

Each new trigger begins beyond every completed stage, so every already configured region lies in every later K_j . The omitted well sum therefore has absolute value and spatial jets through order four bounded by $\sum_j 2^{-j} \leq 1$ there. Its Hessian is also positive semidefinite. Feasibility is proved below. This convention reserves the entire future allowance before selecting any future coefficients. We apply the local all-fiber lemmas to the final twist with these allowances; we do not infer a uniform curvature statement from small perturbations on compact subsets of the total space.

Initially put

$$a = a_\psi = a_{\text{init}} e^t, \quad Y = t, \quad \lambda = p = 1, \quad \kappa = (1 + Y)^{-1}, \quad (8.15)$$

with a_{init} so large that $H \leq \sigma$, and choose B, Ψ with equal initial constants. Up to a positive scale and an additive constant, this initial spatial potential is the exponential radial example of Seshadri [8, Section 3]. Take center ζ_1 and a constant severity sufficiently large for the initial compact region and the tail allowance. The radial curvature formula is strictly positive on nonzero rank-two Hermitian tests at $t = 0$, by its limit or by direct differentiation of $a_{\text{init}} e^t$. Thus the quiet criterion also applies at the spatial origin.

Before every trigger the radial data will have the form

$$\lambda = p = 1, \quad \kappa = \frac{C_\kappa}{1 + Y}, \quad 1 \leq C_\kappa \leq 1 + Y, \quad H \leq \sigma, \quad (8.16)$$

with C_κ constant on that interval. Then $dY/d\nu = C_\kappa Y$, so the cap can be continued through any prescribed finite interval of ν without a pole. All previously generated nonradial twist terms have bounded positive-order Euclidean jets on the entire outer range: use (8.6) and the affine Z tails. Hence a cap can also be continued until $a = e^{u_s}$ dominates any prescribed bound for those old jets.

Choose the next trigger start ν_B with $Y_B = Y(\nu_B) \gg C_\kappa$, $e^{\nu_B} \geq 1$, and severity constant there at a value satisfying

$$E_0 \geq C_\uparrow u_s(\nu_B). \quad (8.17)$$

The reset below provides this inequality and any additional fixed threshold. Put

$$\xi = Y_B(\nu - \nu_B), \quad \nu_* = \nu_B + Y_B^{-1}, \quad R = e^{\nu_*/2}. \quad (8.18)$$

These data precede the choice of c . Retain the capped radial data through

$$l = T_* + 4, \quad T_* = 10\{c + (1 + L_n) \log N\}. \quad (8.19)$$

For large c this lies inside $\nu \leq \nu_B + 2/Y_B$, since $(T_* + 4)/N \rightarrow 0$. Throughout this fixed continuation $Y \asymp Y_B$ and $u_s = u_s(\nu_B) + O(1)$, with constants allowed to depend on the already fixed cap data.

Lemma 8.3. *Before choosing c at stage j , one can fix $D_j \geq j$ such that, for every sufficiently large permitted c and every final choice of the reserved tails,*

$$\ell(s, \zeta_j) \leq D_j - 2 \quad (|s| \leq R e^{c/N}). \quad (8.20)$$

Proof. On the spatial region following the previous center-change interval, the center is ζ_j and $\ell(s, \zeta_j) = B(s)$, including on shear supports. The same identity holds on the initial cap at the first stage. Thus the bound there follows from the fixed cap continuation up to $\nu_B + 2/Y_B$. The earlier spatial region is a fixed compact set K with already chosen profiles and centers; future stages do not change $q(x, s)$ on K . The reservation gives $|\psi - \psi_{\text{old}}| \leq 1$ there, including the current and all later wells. Both centers are in the unit disk, so $|\zeta_j - \zeta(s)| \leq 2$. Choose $A > \log 4 + \sup_K \psi_{\text{old}} + 1$. For each $s \in K$, $F(x, s) \rightarrow \infty$ as $x \rightarrow \infty$. Continuity, a finite cover, and monotonicity in x give a

common finite X with $F(X, s) > A$ on K . Inverting the fiber relation therefore gives $\tau \leq e^X$. Since $q \geq 1$, the radial potential is bounded by $\sup_K B + e^X$; the fixed shear adds at most $2 \sup_K |c_{\text{sh}}|$. At center coincidences this argument is understood by the smooth limit $\tau = 0$. These bounds are independent of c and of the future wells. Add two and increase to at least j . $\square$

We always require

$$c \geq j(D_j + 1) \quad (8.21)$$

in addition to the finitely many current thresholds. Thus $D_j \rightarrow \infty$ and $D_j/c_j \rightarrow 0$.

8.3 The severity trigger

Choose a smooth convex nondecreasing h_b on $(-\infty, 1)$, zero for $\xi \leq 0$, equal to $-\log(1 - \xi)$ up to an affine function near 1. To construct it, cut on the nonnegative second derivative $(1 - \xi)^{-2}$ smoothly and integrate twice from zero. Use

$$E = E_0 + k_b h_b(\xi), \quad Z_{j,\text{in}} = M_b k_b^2 h_b(\xi) \quad (8.22)$$

until severity is near E_* . Smoothly saturate E at E_* over a unit interval of its argument, so that $E = E_*$ by $k_b h_b = E_* - E_0 + 1$. Derivatives of the saturation function in that argument are bounded. Keep the unsaturated convex $Z_{j,\text{in}}$ slightly longer, and once $k_b h_b \geq E_* - E_0 + 2$, cut off its second derivative nonnegatively over half the remaining gap to $\xi = 1$. Continue it affinely in ν thereafter. This is a smooth convex nondecreasing function. Such a function of $\log |s|^2$, zero near the origin, is a smooth radial plurisubharmonic function: its transverse and radial Hessian entries are its first and second ν derivatives divided by t .

Near this cutoff the gap $1 - \xi$ is comparable, with fixed stagewise factors, to $e^{-(E_* - E_0)/k_b}$. The final slope is bounded by

$$C M_b k_b^2 Y_B e^{(E_* - E_0)/k_b}. \quad (8.23)$$

The value there is $M_b k_b (E_* - E_0 + O(1 + k_b))$. Multiplication of (8.23) by the remaining gap to ν_* costs only a fixed $O(M_b k_b^2)$, and its product with $4c/N$ tends to zero. Thus for large c ,

$$Z_{j,\text{in}} \leq 2 M_b k_b E_* \quad (|l| \leq 2c). \quad (8.24)$$

Through saturation, its derivatives and those of E through order four have bounds given by fixed prefactors times $(1 + Y_B)^4 e^{4(E - E_0)/k_b}$. The bound with E_* also holds for positive-order derivatives of the affine tail.

The well satisfies (8.14) after increasing c . Indeed throughout severity saturation,

$$l \leq -C(Y_B, E_0) N e^{-E_*/k_b}. \quad (8.25)$$

The absolute value on the right grows exponentially in E_* , because $\log N \sim \gamma E_*$ and $\gamma > 4/k_b$. Homogeneity makes the polynomial evaluations on the whole inner ball exponentially small in this quantity. Cauchy on a collar of scale R/N gives the same conclusion for any fixed derivative order, with only powers of N as further costs. The logarithm in (8.4) is then close to zero with all these derivatives. We can therefore enforce the bound 2^{-j} for its value and derivatives through order $j + 4$. This decay pays all fixed exponential-in- c costs needed at stage j .

Lemma 8.4. *The trigger satisfies the hypotheses of Proposition 5.2, uniformly for every $\tau \geq 0$. After saturation the same criterion applies through $l = 4c$, including the initial part of the post-test term defined below.*

Proof. For raw estimates use Euclidean axes. Here $t \geq 1$, $Y \asymp Y_B$, and $\log Y_B \leq C u_s(\nu_B)$; the last estimate follows by integrating $d \log Y / du_s = \kappa(1 + Y^{-1}) \leq 2.2$ for $Y \geq 1$. Differentiation of $a' = aY/t$ bounds the required radial metric jets by $e^{C_{\text{rad}} u_s}$. Their inverses cost no more. The radial curvature estimate gives

$$C^V(T, T) \geq c\kappa H |T|_V^2 \geq c(a/t) |T|^2, \quad (8.26)$$

since $V \geq aI$ and $\kappa H = C_\kappa Y / ((1 + Y)ta) \geq c/(ta)$. The initial severity pays these bounds, the old data and the future allowance. The trigger cost e^{4E/k_b} fits the quiet raw exponent. The current well is small before saturation. Also $G \geq V$, since all nonconstant extra twist terms in this interval are plurisubharmonic.

For the covariant condition, write

$$\mathcal{B}(\nu) = B(e^\nu), \quad \mathcal{U}(\nu) = \Psi(e^\nu) + Z_{j,\text{in}}(e^\nu),$$

and use primes for ν derivatives within this proof. On the severity-varying interval,

$$\frac{\mathcal{U}''}{\mathcal{U}'}, \frac{|\mathcal{U}'''}{\mathcal{U}''}, \frac{\mathcal{B}''}{\mathcal{B}'}, \frac{|\mathcal{B}'''}{\mathcal{B}''} \leq C\sqrt{\mathcal{U}''}, \quad (8.27)$$

$$\frac{|E'|}{\mathcal{U}'}, \frac{|E'|}{\sqrt{\mathcal{U}''}}, \frac{|E''|}{\mathcal{U}''}, \frac{|E'''}{(\mathcal{U}'')^{3/2}} \leq \varepsilon_b, \quad (8.28)$$

where ε_b can be made as small as required. Away from the pole, h_b and its relevant derivatives are bounded, whereas $\mathcal{B}' = Y/H$ and $\mathcal{B}'' = Y(1 + Y)/H$. The choice $\sigma M_b k_b^2 \ll 1$ makes these background terms dominate. Near the pole let $D_b = Y_B/(1 - \xi)$. Then

$$Z'_{j,\text{in}} \asymp M_b k_b^2 D_b, \quad Z''_{j,\text{in}} \asymp M_b k_b^2 D_b^2, \quad |Z'''_{j,\text{in}}| \leq C M_b k_b^2 D_b^3,$$

and $|E^{(r)}| \leq C k_b^r D_b^r$ for $1 \leq r \leq 3$, including saturation. These prove (8.27) and make (8.28) small by first enlarging M_b and then decreasing σ . The fixed intervening interval of ξ is covered by the first estimates. We also have $\mathcal{U}'' \geq 1$.

At fixed τ , the radial part of the Schur metric is the Hessian of

$$J_h = \mathcal{B} + \tau(\mathcal{U} - \bar{F}(x, E)),$$

where the bar denotes averaging from zero to τ in the τ variable. Bounded severity derivatives of $\bar{F}$ and (8.28) show that the first two derivatives of $\mathcal{U} - \bar{F}$ are positively comparable with those of $\mathcal{U}$ and its third derivative is bounded by $C(\mathcal{U}'')^{3/2}$. Use affine axes normalized by the Hessian of $\mathcal{U}$, with radial radius $\sqrt{t/\mathcal{U}''}$ and transverse radii $\sqrt{t/\mathcal{U}'}$. At a radial point, the only nonzero holomorphic connection types for a radial Hessian metric are radial–radial to radial and radial–transverse to transverse. Differentiation gives the bounds

$$\frac{|J_h'''/J_h'' - 1|}{\sqrt{\mathcal{U}''}}, \quad \frac{|J_h''/J_h' - 1|}{\sqrt{\mathcal{U}'}}. \quad (8.29)$$

They are bounded by (8.27). No estimate of a radial/transverse condition number is used.

Write the remaining Hessian as τG_e , with $G_e \geq 0$. Its size and first jets in these axes are bounded: old Euclidean bounds are dominated by the large initial a , and the current and future wells have the reserved absolute bounds. The normalizing radii are at most $a^{-1/2}$ because this

background is at least $V \geq aI$. If h_r denotes the radial Schur metric, the exact connection difference is, with the usual index placement,

$$\Gamma^h - \Gamma^{h_r} = \tau(\partial G_e - \Gamma^{h_r} G_e)(h_r + \tau G_e)^{-1}. \quad (8.30)$$

Since $h_r \geq c\tau I$, τh^{-1} is uniformly bounded for all $\tau > 0$. Thus the full connection is uniformly bounded as well.

In these axes the first derivative of E has norm $|E'|/\sqrt{\mathcal{U}''}$; the complex Hessian entries are bounded by $|E'|/\mathcal{U}'$ and $|E''|/\mathcal{U}''$, and the only pure second derivative has size $|E'' - E'|/\mathcal{U}''$. Consequently subtracting the connection times the first derivative gives a small $\nabla_h^{2,0} E$. Since the actual G is at least the normalizing background, the same smallness holds in G units. This proves all quiet smallness conditions uniformly in τ .

After saturation, severity derivatives vanish. Through $l = 4c$ the well's raw costs are bounded by $e^{C(c+\log N)}$; the post term below has cost at most a fixed multiple of $N^4 e^{2m_z c}$. Both fit ϵE_* by (8.9), while the inverse and radial curvature estimates persist. $\square$

8.4 Testing disks and the post ramp

Choose c sufficiently large that the degree in (8.3) meets the grid accuracy required for h_j , and $N > h_j$. At every good lift v , use a smooth cutoff supported in $|s - Rv| < 2RN^{-k_*}$ and equal to one on $|s - Rv| \leq RN^{-k_*}$. Assign on the inner ball

$$c_{\text{sh}} = \alpha b_v, \quad \alpha = \exp(-C_s E_* - C_w d \log N), \quad (8.31)$$

and multiply by the cutoff in the collar. The phases have modulus one, are independently assignable by projective direction, and repeat on every lift of that direction. The outer balls are disjoint. A cutoff $\chi(|s - Rv|^2/(RN^{-k_*})^2)$ has fixed-order multilinear derivative bound $C_r(RN^{-k_*})^{-r}$ with absolute C_r . At most one contributes at any point, so there is no factor for the number of nodes or dimension.

On the derivative supports, the shell estimate gives

$$\psi \geq -2d(k_* + 11) \log N - C_{\text{old}}. \quad (8.32)$$

Old values are bounded on the fixed cap continuation, future values have the allowance, and the trigger term is nonnegative. These balls have $|l| \leq \log N + 1$ and lie beyond severity saturation. First and pure second derivatives of ψ also fit the quiet raw bounds; the extra background first-derivative cost is at most $a\sqrt{t}$. The constants C_s, C_w pay respectively the severity exponent of Proposition 5.3 and half the loss (8.32) together with the four cutoff derivatives. Large c pays fixed old factors. Thus that proposition applies for all fiber radii with metric comparison loss at most $1/2$. Where a shear coefficient is constant, its potential term is pluriharmonic.

At a node the new well is $-2dc + O(1)$; on $|s| = S$ it is at most $2d(c + \log N) + O_d(1)$. All other values on the fixed continuation are bounded independently of c , except the trigger term. The post term is still zero. Hence

$$\psi_{\text{node}} \leq -2dc + 2M_b k_b B_* c + C_{\text{old}}, \quad (8.33)$$

$$\psi_{\text{sphere}} \leq 2d(c + \log N) + 2M_b k_b B_* c + C_{\text{old,d}}. \quad (8.34)$$

For a small fixed profile-dependent $c_0 > 0$, the disk

$$|y - \zeta_j| < c_0 e^{-(\psi + E_*)/2} \quad (8.35)$$

has $\tau \leq e^{-E_*}$ and radial potential increment at most one. Indeed, by the axis estimates in Lemma 5.1, at $\tau = e^{-E_*}$ the normalized relation gives $F = -E_* + O(1)$ uniformly: integrate $1/q - 1 = O(e^{E_*}\tau)$ up to that value of τ . Monotonicity proves the disk assertion.

At a node the holomorphic shear $w \mapsto w + i\alpha b_v(y - \zeta_j)$ has imaginary increment $\text{Re}(\alpha b_v(y - \zeta_j))$, exactly the added potential. Writing $z = y - \zeta_j$, the clearance on the disk (8.35), over any w with $\text{Im } w = D_j$, is therefore

$$\begin{aligned} & \text{Im}(w + i\alpha b_v z) - \ell(Rv, \zeta_j + z) \\ &= D_j - \ell_{\text{rad}}(Rv, \zeta_j + z) \geq D_j - B(Rv) - 1 \geq 1. \end{aligned}$$

The last inequality uses (8.20) on the center axis. Thus the shear consumes none of the height margin, and the sheared disk lies in the tube. Its logarithmic radius is at least

$$(d - M_b k_b B_* - B_*/2)c - C_{\text{old}} \geq (13d/16)c - C_{\text{old}},$$

which exceeds $m_0 c$ for large c . On the sphere there is no shear, and Cauchy's inequality on (8.35) bounds the transverse derivative of a function of modulus at most one by

$$\exp\{[d(1 + \gamma B_*) + M_b k_b B_* + B_*/2]c + C_{\text{old},d}\} \leq e^{p_0 c}.$$

Also $\alpha^{-1} \leq e^{p_1 c}$. These are precisely the constants (8.12), with strict margins to absorb the prefactors. Choose the phases supplied by Proposition 3.5. All geometric estimates hold for arbitrary phases.

For the post ramp, add a smooth convex nondecreasing function $Z_{j,\text{out}}$ of l , zero through $2c$, and equal to $e^{m_z(l-2c)}$ up to affine terms from $2c + 2$ onward. This is obtained by cutting on its positive second derivative and integrating twice. Continue through $T_* + 2$, then cut off that second derivative nonnegatively so that the function is affine after $T_* + 3$. Independently turn on the severity slope B_* in l over $[4c, 4c + 1]$, and turn it off over $[T_*, T_* + 1]$. Write E_{post} for the final value. For large c ,

$$E \geq B_*(c + l)/10 \quad (2c \leq l \leq T_* + 4), \quad E_{\text{post}} \geq B_* T_*/2. \quad (8.36)$$

Lemma 8.5. *The quiet criterion holds throughout the post ramp. Beyond $l = T_* + 4$, the sum of all already generated additive twist terms other than Ψ and the constant offsets, and each such term individually, has Euclidean positive-order jets through order four bounded on the whole outer range by*

$$D_{\text{out}} = \exp(\epsilon_1 E_{\text{post}}/3). \quad (8.37)$$

Proof. Each derivative of a function of l costs a fixed power of N . Thus $Z_{j,\text{out}}$ has positive-order bounds through order four of size $CN^4 e^{m_z \max(0, l-2c)}$. These and (8.5) fit the raw quiet exponent by (8.9) and (8.36). Other radial and old-data bounds remain fixed on the cap continuation. Through $l = 4c$, severity is constant and the preceding lemma applies.

Where severity varies, take $\mathcal{U} = \Psi + Z_{j,\text{out}}$. Before division by t , its transverse and radial Hessian entries have sizes

$$m_z N e^{m_z(l-2c)}, \quad (m_z N)^2 e^{m_z(l-2c)}.$$

For $l \geq 4c$, $m_z(l - 2c)$ exceeds $C_{\text{pol}}(l + c + \log N)$ by a fixed positive multiple of $c + l$. Thus this Hessian dominates the current well and its first Hessian jets in normalized axes. The affine $Z_{j,\text{in}}$ tail has coefficient at most a fixed factor times $e^{E_*/k_b} \ll N$; old terms and future tails are covered as before. Moreover $\mathcal{U}''/\mathcal{U}', |\mathcal{U}'''|/\mathcal{U}'' \leq C m_z N$, while $\sqrt{\mathcal{U}''} \gtrsim m_z N e^{m_z(l-2c)/2}$. Severity derivatives

of orders $r \leq 3$ are at most CB_*N^r . Thus the connection ratios hold and all severity ratios are small. Equations (8.29) and (8.30) again prove the pure Hessian condition uniformly for all τ . Every extra Hessian is positive semidefinite in this interval.

Past $T_* + 4$ the new well satisfies (8.6), and the Z terms are affine in ν . Their positive-order Euclidean jets are bounded on the entire outer range by their endpoint bounds up to fixed constants. Their logarithmic costs are at most $C(1 + m_z)T_* + 4 \log N + C_{\text{old}}$ with an absolute C . The absolute constant C_* in (8.9) was chosen to pay this inequality too. Equation (8.36) then implies (8.37), after increasing c for the finite sum of old bounds, the number of generated terms, and dimensional prefactors. The dimension-dependent L_n multiplies T_* on both sides and changes no leading comparison. $\square$

8.5 Acceleration, center change, and reset

At constant E_{post} , increase κ from the cap value to $\kappa_* = 1.08$ on a fixed-factor interval of Y after $l = T_* + 4$, keeping $\lambda = p = 1$. The transition can preserve $Y\kappa_Y \geq -\kappa$: multiply the difference between the cap value and κ_* by a nondecreasing cutoff in $\log Y$. The upward derivative only improves that inequality. The u_s increment is bounded in terms of the fixed prior Y_B , independently of large c . Keep $\kappa = \kappa_*$ until

$$u_s = \epsilon_1 E_{\text{post}}. \quad (8.38)$$

For large c this endpoint lies after the transition and has $H \geq \sigma$. Indeed, with $p = 1$, $d \log H / du_s = (\kappa_* - 1)(1 + Y^{-1}) \geq .08$, whereas the initial values at the end of the cap have fixed stagewise bounds as $c \rightarrow \infty$.

Until (8.38), apply the quiet criterion. The radial raw jets cost $e^{C_{\text{rad}} u_s}$, the extra twist jets are bounded by (8.37), and the curvature lower bound follows as in (8.26). Severity derivatives vanish. The choices of ϵ_1 and B_* pay all these estimates. At entry $a_\psi = e^{\epsilon_1 E_{\text{post}}}$. By the scaled-axis estimate of the radial lemmas, an extra Euclidean Hessian with two further jets bounded by D costs at most $CD/(a_\psi H)$ in scaled axes. Consequently at entry and through the later aligned intervals,

$$|j^{\leq 2} G_e|_{\text{scaled}} \leq \delta/H. \quad (8.39)$$

Here we leave a fixed fraction of the δ budget for future wells. The ratio D_{out}/a_ψ is exponentially small at entry and only decreases later, since a_ψ increases. During an actual profile switch its new offset is accounted for by Proposition 7.1; outside the switch, that offset is constant.

Lemma 8.6. *At constant E_{post} the center can be changed from ζ_j to ζ_{j+1} , and the data can then be restored to*

$$\lambda = p = 1, \quad \kappa = 1, \quad H = \sigma,$$

using finite clock intervals and only the aligned and center-change regimes. The old-twist bounds needed for choosing the profile shift are available independently of the targeted λ, H .

Proof. Initially grow H with $\kappa = \kappa_*$ and $p = \lambda = 1$. Use the $\lambda = 1$, $H \geq \sigma$ alternative of Proposition 6.4. Ultimately we will lower p slightly below one for a bump, thereby increasing λ , and then restore $p = 1$ and return smoothly to $\kappa = 1$.

We first justify the order in which the targets are chosen. During the growth and the bump, $p \leq 1$ and $\kappa = \kappa_*$, so $d \log H / du_s \geq .08$. Thus

$$\int_{u_{\text{entry}}}^{\infty} \frac{du_s}{H} \leq \frac{1}{.08 H_{\text{entry}}}.$$

This bounds the total increase of Ψ before the final transition independently of the size of either target. The transition from κ_* to one takes a fixed $\log Y$ interval and leaves H nondecreasing, adding another fixed bound to Ψ . Moreover $\kappa \geq 1$ gives $d \log Y / du_s \geq 1$, hence

$$0 \leq \nu - \nu_{\text{entry}} = \int \frac{du_s}{Y} \leq Y_{\text{entry}}^{-1}$$

through the growth and holding intervals. The old nonradial twists therefore have bounded values on one fixed compact spatial interval, independently of the targets. Their scaled derivatives are bounded by the fixed Euclidean bounds, since the scaled coordinate radii are at most one. Derivatives of Ψ in these axes are $O(1/H) \leq O(1/\sigma)$. The forward switch and center move each occupy a fixed amount of the variable Ψ ; these amounts can be included in the same bounds. Include the reserved future value and jet allowance as well. These are exactly old-twist bounds independent of λ, H , as required by Proposition 7.1.

Choose its shift L_0 from these bounds and E_{post} . This fixes its profile constants C_f and a sufficient target λ , and that target determines a sufficient target H . This is the order of parameter selection in Proposition 7.1. The radial path reaches the targets in the other order: grow H first until it exceeds every required multiple of

$$C_{E_{\text{post}}}(1 + \log \lambda)^2 \quad \text{and} \quad C_f(1 + \log \lambda)^2.$$

Choose a smooth bump of $1 - p \geq 0$ with small amplitude and small derivatives in $\log Y$, and with integral in u_s equal to the desired $\log \lambda$. Such a bump exists by making its middle plateau arbitrarily long; the endpoint ramps can have fixed small amplitude and sufficiently long fixed length, while the plateau length supplies the remaining integral. The conversion $d \log Y / du_s = \kappa(1 + Y^{-1})$ and its bounded derivatives show that bounds through order two in $\log Y$ follow from the same bounds in u_s . Throughout this bump H increases, and the already chosen lower bound for H pays the entire range $0 \leq \log \lambda(s) \leq \log \lambda$. Apply the large- H alternative of Proposition 6.4.

Restore $p = 1$ and lower κ smoothly to one. The downward transition can be made slowly enough in $\log Y$ that $Y\kappa_Y \geq -\kappa$, with absolute derivative bounds. Since $\kappa \geq 1$ here, H stays large. Now $\kappa = p = 1$, so both H and λ are constant. Perform the forward profile switch, the center move, and the reverse switch of Proposition 7.1. Interpolate the two center values by a flat-ended convex combination, keeping the center in the unit disk. The old bounds and targets have been chosen in the order just proved. The changes take a bounded interval of Ψ , hence a finite interval of u_s because $d\Psi/du_s = 1/H$ and H is fixed, even though this interval can be very long. Restore q_0 and retain the new constant offset. The center is now ζ_{j+1} . The offset may be arbitrarily large, but it is now constant and changes no derivative of the twist. Its value becomes part of the old data when the next testing height is chosen, before the next accuracy parameter c .

Raise κ to κ_* again and use a bump with $1 < p < \kappa_*$ to bring $\log \lambda$ down to zero. A small positive plateau and fixed ramps give any prescribed finite integral of $p - 1$; stop precisely when that integral equals the existing $\log \lambda$. Thus λ never falls below one. Also

$$\frac{d \log H}{du_s} = \kappa_* - p + \frac{\kappa_* - 1}{Y} > 0,$$

so the large- H aligned condition remains valid. Restore $p = 1$.

Finally reduce H to σ . Transition κ to $1/2$, hold there for an adjustable interval, then return to one from below. For $p = 1$ the exact relation is

$$\frac{d \log H}{d \log Y} = 1 - \kappa^{-1}. \tag{8.40}$$

Its value is -1 on the middle plateau. The endpoint transitions have fixed finite effects on $\log H$; the initial H can be made as large as needed. Choose the plateau length so that the final value is exactly σ . Throughout this descent $H \geq \sigma$, $\lambda = 1$, and $1/2 \leq \kappa \leq \kappa_*$, so the $\lambda = 1$ aligned alternative applies. Choose the downward transitions slowly enough to preserve the same derivative inequality as above.

All these intervals have finite u_s or finite $\log Y$ length. Equations (8.1) show that Y remains finite on each finite u_s interval and that the radius advances by a finite positive amount. A putative pole of an indefinitely continued accelerated equation is therefore never reached during an episode. This proves the claim. $\square$

Lemma 8.7. *From the endpoint in Lemma 8.6, severity can be increased and a new cap started so that the next trigger has (8.17), while the metric continues to satisfy the quiet criterion. The cap can be extended to arbitrarily large spatial radii.*

Proof. The spatial scale u_s may now be enormous, and E_{post} need not dominate it. Proposition 5.2 permits any fixed holomorphic affine frame at the evaluation point. We first use V -unit affine axes, where the curvature scale is the fixed σ and the relevant jets are bounded independently of u_s . This will justify the initial severity increase before arranging the size inequality needed for the next trigger.

First hold $\kappa = p = \lambda = 1$ and $H = \sigma$. Let E increase smoothly as a function of u_s , starting at E_{post} , with a fixed flat-ended transition to slope $4C_{\uparrow}$. Its derivatives through order four in u_s are bounded by constants depending on the fixed parameters, not on σ or the possibly enormous u_s at entry. Continue long enough that

$$E \geq 2C_{\uparrow}u_s \quad (8.41)$$

and any desired fixed threshold. The greater slope ensures that this occurs after a finite interval.

We verify the quiet criterion even before (8.41). Use V -unit affine axes. The scaled-jet radial estimates give metric jets of orders $r = 1, 2$ bounded by $CH^{r/2}$ in these axes, and scaled derivatives of u_s through order four are bounded by an absolute constant. Therefore

$$|\partial^{[r]}E|_V \leq CH^{r/2}, \quad 1 \leq r \leq 4.$$

The extras in (8.39) satisfy the same bounds after this normalization, and $G \geq V$. For σ small, the severity terms in $\mathcal{N}$ are small relative to G , so $h \asymp (1 + \tau)V$, with first derivatives bounded in these axes by $C(1 + \tau)\sqrt{H}$. Its connection is consequently $O(\sqrt{H})$ uniformly in τ . The first, complex second, and pure covariant second severity derivatives are small for every τ .

The raw curvature margin on this hold is $C^V(T, T) \geq c\sigma|T|^2$ in V -unit axes, because $\kappa = 1$ and $H = \sigma$. All other raw data are bounded there by fixed constants. The previously chosen E_{post} exceeds the fixed profile and σ thresholds, so it pays this margin even when it is much smaller than the current u_s . Thus there is no gap before severity catches up.

After (8.41), keep the same slope while turning κ into a cap. Here is an explicit admissible transition. If Y_0 is its starting clock value, set

$$A(Y) = 1 + Y_0 + \int_{Y_0}^Y \eta_0(v/Y_0) dv, \quad \kappa(Y) = \frac{A(Y)}{1 + Y},$$

where η_0 is smooth, equals one near 1, decreases nonnegatively to zero, and is zero for arguments at least two. For $Y \leq Y_0$ use $A = 1 + Y$. Then $1 \leq A \leq 1 + Y$, so $(1 + Y)^{-1} \leq \kappa \leq 1$. Moreover

$$Y\kappa_Y + \kappa = \frac{YA'(Y)}{1 + Y} + \frac{A(Y)}{(1 + Y)^2} \geq 0.$$

The first two derivatives of κ in $\log Y$ have absolute bounds, because Y/Y_0 ranges over a fixed interval while η_0 changes. For example, writing $b = Y/(1+Y)$ and $D = Y\partial_Y$, one has $D\kappa = b(\eta_0 - \kappa)$ and $D^2\kappa = b(1-2b)(\eta_0 - \kappa) + bD\eta_0$. At the end A is the new constant C_κ . Since $\kappa \leq 1$ and $p = 1$, H is nonincreasing and stays at most σ .

The same normalized jet and connection estimates continue through the transition and the cap. To check the raw curvature margin there, note that $Y \geq t$: it holds initially, and $d\log Y/d\nu = \kappa(1+Y) \geq 1$ preserves it. Also $u_s \geq t$ after choosing the initial constant large enough, since $du_s/dt = Y/t \geq 1$. Hence

$$\kappa H \geq c/(ta) \geq ce^{-2u_s} \quad (t \geq 1),$$

and (8.41) with large $C_\dagger$ pays the quiet lower bound $e^{-\epsilon E}$. The positive-order extras stay bounded relative to V on an arbitrarily long cap. More explicitly, a fixed Euclidean jet bound D_{out} costs at most $C(D_{\text{out}}/a)H^{r/2}$ in V -unit axes for Hessian jets of order $r \leq 2$, since $aH = Y/t \geq 1$. The ratio only decreases as a grows. Future wells are included in the fixed allowance until their own activations.

Coast as far in ν as desired. Shortly before the next chosen trigger, turn off the severity slope smoothly over a fixed-length u_s interval. The margin in (8.41) is sufficient to retain $E_0 \geq C_\dagger u_s(\nu_B)$ after this bounded final increment, by making the cap endpoint large enough. The existing raw jet bounds can also be absorbed by making that endpoint large. This produces (8.17) and the next cap data. $\square$

8.6 Global realization and completeness

We now carry out the recursion. At each stage first extend the cap so that its trigger starts beyond all previous stages and with $\nu_B \geq j$. Fix its old-data bounds and testing height by Lemma 8.3. Then choose c large enough for (8.21), the polynomial approximation for h_j , (8.14), and all the finite thresholds in the trigger, test, post ramp, and aligned entry. The later finite center-change targets are chosen in the order proved in Lemma 8.6. Lemma 8.7 completes the cycle. Thus no clock or already configured profile is determined by an unknown future well.

For completeness, the future allowance is valid in every kind of local regime. In a quiet interval, later wells have positive-semidefinite Hessians and bounded absolute jets; they preserve $G \geq V$, and the preceding connection proof uses exactly these bounds. In an aligned or switch interval, their scaled Hessian jets are bounded by $C/(a_\psi H)$, which fits the reserved part of δ/H at the large chosen a_ψ . In a center move, their values change the radius estimates by at most a fixed factor, and their scaled derivatives are within the old-data allowance included before choosing the shift. On a shear collar they lower the twist by at most one, already included in the lower bound used for its coefficient. Consequently the local lemmas apply to the final infinite twist for every fiber radius, even though curvature margins need not have a positive infimum over all fiber radii.

The condition $\nu_B \geq j$ rules out finite accumulation of stages. Any compact spatial set lies before the activation of all sufficiently late wells. For a fixed derivative order r , those later wells satisfy $|\partial^{[r]}W_j| \leq 2^{-j}$ once $j+4 \geq r$. The Weierstrass criterion in every derivative order proves compact-smooth convergence of $\sum_j W_j$. The Z terms and nonconstant offsets are locally finite. Flat-ended transitions, finite positive clock evolutions, and the prescribed initial data therefore give global smooth spatial data. On a compact spatial set the profile is smooth to $\tau = 0$ with $q(0, s) = 1$; its normalized $F - x$ is smooth there as well. The smooth inversion at the fiber center therefore gives a global smooth potential

$$\ell(s, y) = \ell_{\text{rad}}(s, y) + \text{Re}(c_{\text{sh}}(s)(y - \zeta(s))).$$

The shears are absent where the center changes. All local curvature and positivity conclusions apply, including the fiber axes by their smooth limits.

We spell out the metric comparison used for completeness. In the constant-center regimes the Schur metric is bounded below by a fixed positive multiple of V ; during profile switches, positivity of $\mathcal{N}$ for all τ also gives $h = V + \int_0^\tau \mathcal{N} dv \geq V$. The shear and moving-center comparisons can each be chosen to lose at most a factor $1/2$. Their spatial supports are separated, so these losses do not multiply over infinitely many stages at a point. Hence a single $c_1 > 0$ satisfies

$$g_\ell(\dot{s}, \dot{y}) \geq c_1 V(\dot{s}, \dot{s}) \geq c_1 a_{\text{init}} |\dot{s}|^2 \quad (8.42)$$

everywhere.

A finite-length curve therefore has finite Euclidean spatial variation and remains in a compact spatial set. On such a compact set, for large $|y|$, the metric is also bounded below by a fixed positive multiple of the frozen log-block metric at the point. The tilt coefficients are uniformly bounded in τ there. Indeed they are spatial derivatives of $\psi - F$: on quiet intervals they use bounded F_E ; on switch intervals they use bounded F_β , which is constant in x past the fixed switch cutoff; and on a center move the profile is spatially fixed apart from the already bounded twist. There are only finitely many profile episodes on the compact set, and the well sum is smooth. Thus some compact-set constant C_K bounds the tilt one-form.

Also $q \geq 1$ and $F \leq x$, so at sufficiently large $|y|$ the fiber value satisfies $\tau \geq 1$ and $P = \tau q \geq 1$, uniformly on the compact spatial set. The frozen fiber differential is $dy/(y - \zeta(s))$; freezing means that the center has its point value in this comparison. The block bound and the triangle inequality imply along the curve, when $|y|$ is large,

$$\frac{|dy|}{|y - \zeta(s)|} \leq C_K ds_{g_\ell} + C_K |ds|.$$

Both terms on the right have finite integral by (8.42). Since $|\zeta(s)| < 1$, this bounds the total variation of $\log |y|$ outside a fixed disk and prevents fiber escape. Thus no divergent curve has finite length. Equivalently, a Cauchy sequence lies in a compact Euclidean coordinate set, where the smooth positive metric gives convergence. The base metric is complete.

Proof of Proposition 8.1. The recursion gives a smooth strictly plurisubharmonic potential on $\mathbb{C}^{n+1}$, with nonpositive real sectional curvature by the local lemmas and completeness by the preceding argument. Every required pair (ζ, h) is tested infinitely often. The stages satisfy $R \geq 1$, $N \rightarrow \infty$, $c \rightarrow \infty$, $\log N \leq .005c < c/100$, $D_j \rightarrow \infty$, and $D_j/c_j \rightarrow 0$. Lemma 8.3 supplies the height bound. The node disks, sphere derivative bound, and shear amplitude have the fixed exponents (8.12). Their phases were chosen as permitted by Proposition 3.5, and their geometric realization allows all those choices. That proposition proves the asserted Liouville property. $\square$

Remark 8.8. The potential is necessarily unbounded below, and this can also be seen directly from the disks. At a testing node choose $y - \zeta_j = -\frac{1}{2}e^{m_0 c} \overline{b_v}$. The radial increment is at most one, whereas the shear contributes $-\frac{1}{2}\alpha e^{m_0 c}$. Thus

$$\ell(s, y) \leq D_j + 1 - \frac{1}{2}e^{(m_0 - p_1)c} \longrightarrow -\infty.$$

Here $m_0 > p_1$ and $c/(D_j + 1) \rightarrow \infty$. In particular the elementary holomorphic functions e^{ikw} , $k > 0$, are unbounded on the tube; they do not contradict its bounded-function conclusion.

Completion of the proof of Theorem 1.1. Apply Lemma 2.1 to the base just constructed. Its tube is connected, complete, and diffeomorphic to Euclidean space, hence simply connected. Its real sectional curvatures are at most -1 by that lemma, and its bounded holomorphic functions are constant by Proposition 8.1. The complex dimension is the fixed finite $m = n + 2$.

There is no finite lower sectional-curvature bound. At $s = 0, y = \zeta_1$, the center and profile are constant nearby and there is no shear. On the center axis the spatial curvature tensor of the base equals that of V : the added radial-fiber potential vanishes to second order in $y - \zeta_1$, and its cubic derivatives with two spatial indices and one fiber index vanish there. The initial radial metric has a strictly negative holomorphic-plane curvature in a spatial direction. Corollary 2.2 therefore shows that the tube's sectional curvatures are unbounded below. This proves every assertion of Theorem 1.1. $\square$

The two-sided question. Neither this example nor the independent pinched threefold of OpenAI [7, Theorem 1.1] decides whether a nonconstant bounded holomorphic function must exist under finite two-sided negative real sectional pinching. The present curvature is unbounded below, while the threefold retains nonconstant bounded base-coordinate functions.

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